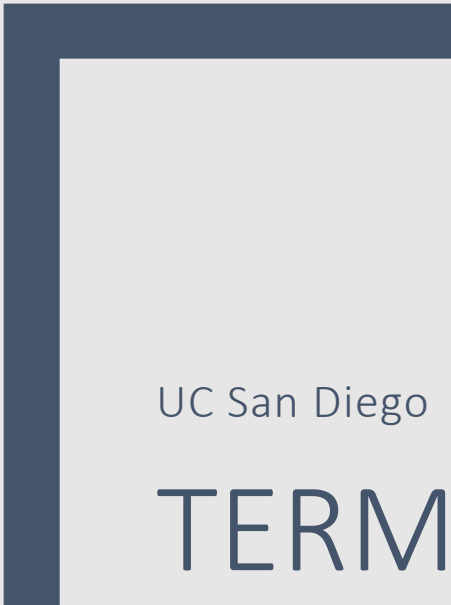




UC San Diego | SE 221

TERM PROJECT

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Abstract

Analysis of a multi-span bridge consisting of 3 frames which are not connect to one another is performed. Two different ground motions, the 1940 El Centro earthquake recording as well as the 1994 Rinaldi record from the Northridge earthquake are used for analysis of the multi-span bridge. Frame 2 is the primary focus, where the two-columns of the asymmetric girder are to be designed appropriately. Assumptions such as the girder being rigid in bending as well as elastic-perfectly plastic behavior is used throughout the design process. Code requirements such as ASCE 7 is utilized throughout this process. It is observed that the yield displacement along with the maximum displacement are 3 and 15 inches respectively. The damping is said to be 5%.

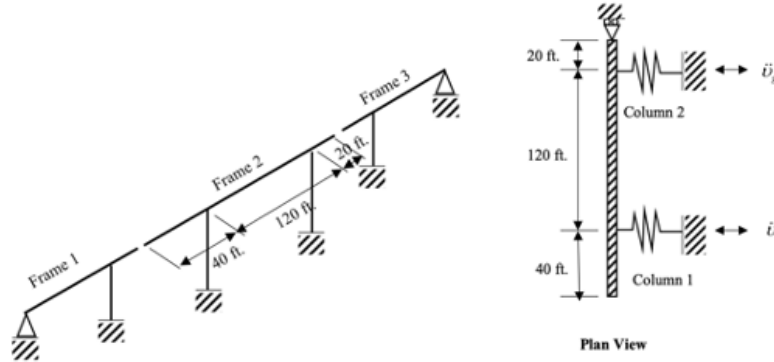


Figure 1: Bridge Structure

Problem 1

The required elastic stiffness along with the required yield strength are to be obtained for the two columns of frame 2, so that the displacement at the top of the column will not exceed 15 inches. In this instance a single degree of freedom (SDOF) approach may be used for the design, along with the equal displacement assumption.

Using the equal displacement method along with the given parameters of S_{DS} , S_{D1} and T_L , a design response spectrum can be generated. The table below shows the following parameters used to generate the response spectrum as well as values used for design in problem 1. The design spectrum shown in the figure below is generated using a piecewise function according to ASCE 7.

Table 1: Parameters for Design Spectrum

Parameter	Value	Unit
S_{DS}	579.13	in/s ²
S_{D1}	386.09	in/s ²
T_L	12	Seconds
T_s	0.67	Seconds
μ	5	-
g	386.09	in/s ²
M	4.19	K*s ² / in
ξ	5	%
W_{total}	1620	Kips

The total design base shear can then be computed using the response spectrum. A vector of design base shear values is established based on the total weight of the structure along with the maximum permitted ductility. The following equation is used to find the total design base shear.

$$V = \frac{W_{total} * S_a}{g * M} \dots (1)$$

From the vector of the total design base shears, the required stiffness of the columns can be obtained using the relationship of $K = V/u_y$ where u_y is the yield displacement. The elastic period can be obtained via an iterative process in which the required yield stiffness and the mass per unit length of the structure are used. Throughout numerous iterations in computer software the elastic period was determined to be 1.534 seconds. Finally, using the geometry of the structure, static analysis can be performed to determine the tributary lengths along with the strength yield relationship to determine which column will be stiffer. These relationships are used to prevent rotation of the frame. The tributary length calculations are shown below.

$$\text{Column 1} = \frac{180 \text{ ft}}{2} - 20 \text{ ft} = \mathbf{70 \text{ ft}} \dots (2)$$

$$\text{Column 2} = \frac{180 \text{ ft}}{2} - 40 \text{ ft} = \mathbf{50 \text{ ft}} \dots (3)$$

The tributary mass relationship is obtained by dividing each tributary length by 2. From these relations, the yield strengths (r) along with the required stiffness (K) of each column are obtained using the equations shown below. A summary of the values obtained for the parameters in problem 1 are shown in the table below.

$$r_y = r_{trib} K_{eq} u_y \dots (5)$$

$$K_{req} = \frac{r_y}{u_y} \dots (6)$$

Table 1.2 : Summary of Problem 1 Parameters

Parameter	Value	Unit
V_{max}	486	Kips
K_{eq}	70.410	Kip/in
r_{y1}	123.225	Kip
r_{y2}	88.018	Kip
K_{col1}	41.075	Kip/in
K_{col2}	29.339	Kip/in
$T_{elastic}$	1.534	Seconds

Problem 2

The columns are now redesigned using the displacement-based method. An equivalent linear elastic system is used to determine the load displacement relationship of each column. The damping ratio is 5%. The design spectrum in figure 2 can be modified to develop a new displacement response spectrum. This response spectrum can be obtained using the following formula.

$$S_d(T, \xi_{eq}) = S_d(T, \xi = 5\%) \sqrt{\frac{0.07}{0.02 + \xi_{eq}}} \dots (7)$$

The equivalent stiffness can be obtained by finding the area under the curve of one linear cycle. This is obtained using the formula in equation 8. Plugging equation 8 into equation 9, the equivalent damping can be obtained.

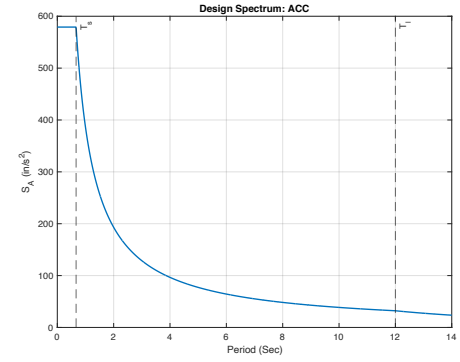


Figure 2: Design Spectrum

$$EH = 2r_y(u_{max} - u_y)2 \dots (8)$$

$$\xi_{eq} = \frac{EH}{2\pi u_{max}^2 K_{eq}} \dots (9)$$

The equivalent damping term using the relations above is 0.5093. Plotting the modified response spectrum taking into account the equivalent damping term the following displacement plot is generated (figure 3).

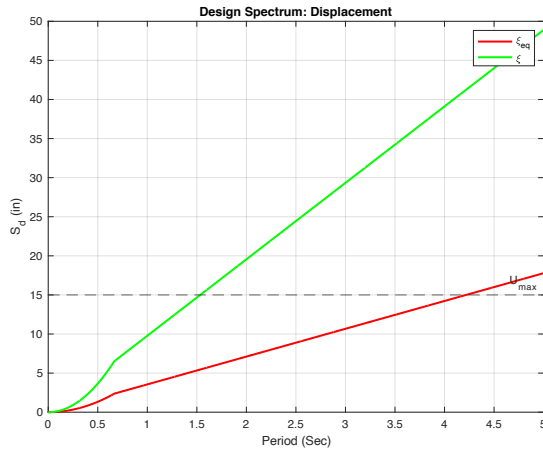


Figure 3: Design Response Spectrum (Displacement)

Using the design response spectrum in figure 3, the location of the equivalent period can be obtained. In this case, the fundamental period is located on the equivalent damping line where it is equal to the maximum allowable displacement ($U_{max} = 15''$). At this location it was deemed that the equivalent period is 4.219 seconds. From this equivalent period, the equivalent stiffness can be found. The equivalent stiffness and equivalent shear are found using the following formulas:

$$K_{eq} = \frac{4\pi^2 M}{T_{eq}^2} = 9.306 \frac{k}{in} \dots (10)$$

From the equivalent stiffness, the equivalent shear can be obtained by $K_{eq} U_{max}$. The equivalent shear is 139.59 kips. From this shear a “yield stiffness” can be obtained where the stiffness uses the same relation as above and is 46.531 kip/in. From these parameters the elastic period can be computed with the same formula as that of problem 1. The elastic period for problem 2, a two DOF analysis, is 1.887 seconds. From all of the values above, the required stiffness and yield strengths can be obtained for each column. The same tributary length and mass relationships as described in problem 1 are used (same formulas) and the values for each column can be seen in the table below. Using the tributary length relationship as discussed in problem 1, the first column is required to be 1.4 times stiffer than column 2.

Table 2.1 : Summary of Problem 2 Parameters

Parameter	Value	Unit
K_{col1}	27.143	Kip/in
K_{col2}	19.388	Kip/in
r_{y1}	81.429	Kip
r_{y2}	58.164	Kip
K_{eq}	9.306	Kip/in
K_{eq_yield}	46.531	Kip/in
$T_{elastic}$	1.887	Seconds
ξ_{eq}	0.509	-

Problem 3

The code for problem 3 is attached to the separate file. A Modified Newton Raphson iteration with constant average acceleration method is used to determine the restoring forces along with the displacements of the structures in problems 1 and 2. For this section Dr. Shings’ lecture 19 notes “Nonlinear Time Integration Methods” is utilized.

Problem 4

The design of the structure in problems 1 and 2 are ran with the code in problem 3 to determine the time history response. Problem 1 is a SDOF whereas problem 2 is a 2DOF system. Matrices will be used for analysis and the derivation of each matrix can be seen below. The determination of the fundamental period is essential for all parts of this problem. The fundamental period is determined by taking the average of the two elastic periods computed in problems 1 and 2.

$$T_{fund} = \frac{T_{elastic1} + T_{elastic2}}{2} = \mathbf{1.7103 \text{ sec} \dots (11)}$$

Mass Matrix

The mass matrix is formulated similar to the direct stiffness assembly method. A unit acceleration is applied in the direction of each DOF, using statics and the principal of equilibrium, the respective entries of each index in the 2x2 mass matrix is obtained.

Analysis is shown in figure 4. The mass matrix for all parts of the analysis of the multi-span bridge will remain the same. The units of the mass matrix are kip*s²/in.

$$\tilde{M} = \begin{bmatrix} 19M & M \\ 36 & 18 \\ M & 13M \\ 18 & 36 \end{bmatrix}$$

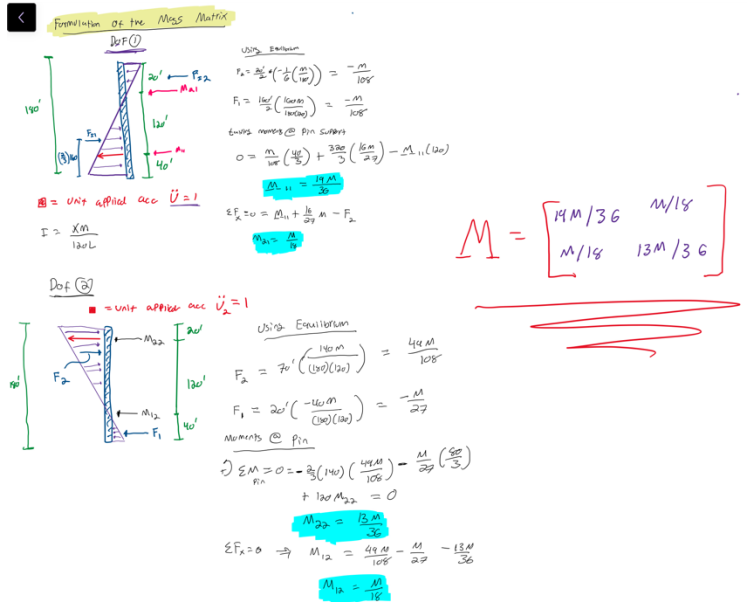


Figure 4: Formulation of Mass Matrix

Stiffness Matrix

The stiffness matrix will vary based on the structures designed in problem1 and problem 2, using the two different methods. The stiffness matrices are uncoupled and can be seen below for each problem where the units are kip/in for each index.

$$\widetilde{K}_{prob1} = \begin{bmatrix} 41.075 & 0 \\ 0 & 29.229 \end{bmatrix} \quad \widetilde{K}_{prob2} = \begin{bmatrix} 27.143 & 0 \\ 0 & 19.388 \end{bmatrix}$$

Damping Matrix

The damping matrix is formulated using the Rayleigh damping procedure. The eigenvalue problem is solved, and the natural frequencies are obtained. From these natural frequencies, the Rayleigh damping matrix for each structure designed in problems 1 and 2 can be formulated. The natural frequencies for problem 1's system are 4.0965 and 4.6641 rad/s and the natural frequencies for problem 2's system are 3.3301 and 3.7915 rad/s. From these values the damping matrices can be formed.

$$\widetilde{C}_{prob1} = \begin{bmatrix} 0.95184 & 0.05084 \\ 0.05084 & 0.66536 \end{bmatrix} \quad \widetilde{C}_{prob2} = \begin{bmatrix} 0.79281 & 0.05084 \\ 0.05084 & 0.55176 \end{bmatrix}$$

PART A

The 1940 El Centro recording is used in this part with a scale factor so that the pseudo spectral acceleration at the fundamental period if the structure is equal to that of the design spectrum. The scaling factor is obtained using a SDOF analysis in which the fundamental period is used, and any arbitrary mass or stiffness can be found so long as the period is equal to the fundamental period. A linear time history analysis is performed, and the scaling factor is obtained by dividing the acceleration at the fundamental period by the acceleration according to the design spectrum.

$$SF = \frac{S_{a_fund}}{S_{a_SDOF}} = 3.6654 \dots (12)$$

The response of the structures designed in problems 1 and 2 can be seen in the following figures. For the structure designed in problem 1, the maximum displacement observed is 17.824 inches for both columns and the maximum restoring force was seen to be 123.23 kips for column 1 and 88.018 kips for column 2 in structure 1. Whereas looking at structure 2, the maximum displacement is 16.963 inches for both columns and the maximum restoring force for columns 1 and 2 are 81.429 kips and 58.164 kips respectively. The parameters are summarized in the table 4.1.

Table 4.1: Response Values Part A		
Parameter	Value	Unit
Max Displacement (Structure 1)	17.8240	in
Max restoring force (Structure 1,Col1)	123.23	kip
Max restoring force (Structure 1,Col2)	88.018	kip
Max Displacement (Structure 2)	16.9625	in
Max restoring force (Structure 2,col1)	81.429	Kip
Max restoring force (Structure 2,col2)	58.164	Kip

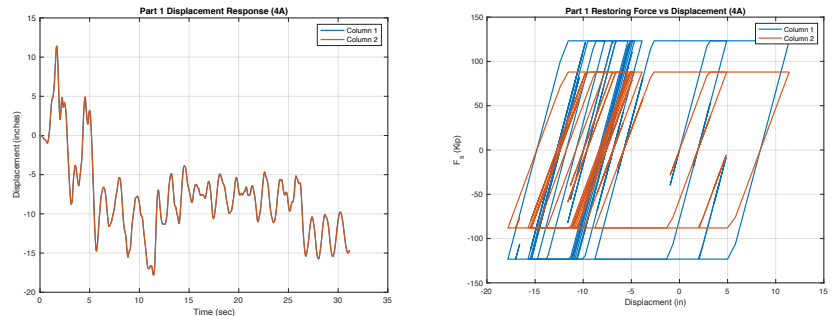


Figure 5: Structure 1 Response (El Centro, 4A)

As observed, structure 2 experiences a smaller maximum displacement. The restoring force of each column in structure 2 shows that column 1 has a larger restoring force compared to column 2. Although both columns in structure 2 experience a smaller restoring force compared to those in structure 1. Structure 1 shows the largest restoring force for each column.

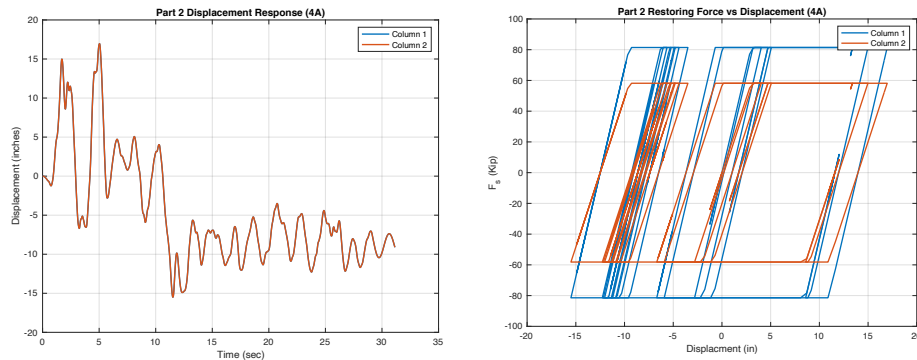


Figure 6: Structure 2 Response (El Centro, 4A)

Part B

The same procedure shown in part a is used to determine the scaling factor. Part B now multiplies the fundamental acceleration by a factor of 2. This will significantly increase the scaling factor.

$$SF = \frac{2S_{a_fund}}{S_{a_SDOF}} = 7.3307 \dots (13)$$

The response values of both structures subjected to the El Centro recording with a scaling factor that correspond to 2 times that of the design spectrum can be seen in table 4.2.

Table 4.2: Response Values Part B		
Parameter	Value	Unit
Max Displacement (Structure 1, both col)	48.718	in
Max restoring force (Structure 1,Col1)	123.23	kip
Max restoring force (Structure 1,Col2)	88.018	kip
Max Displacement (Structure 2)	52.26	in
Max restoring force (Structure 2,col1)	81.429	Kip
Max restoring force (Structure 2,col2)	58.164	Kip

As seen in the responses, the maximum displacements significantly increase due to this scaling factor. For both structure, both columns behaved the same, with structure 1 having a maximum displacement of 48.718 inches and structure 2 being 52.26 inches. The restoring forces remain the same as that in part A where the maximum restoring force for columns 1 and 2 are greater in structure 1 compared to structure 2. Structure 1 has restoring forces almost 1.5 times larger for column 1 and column 2 compared to structure 2. The response for structure 1 can be seen below (displacement and restoring force response).

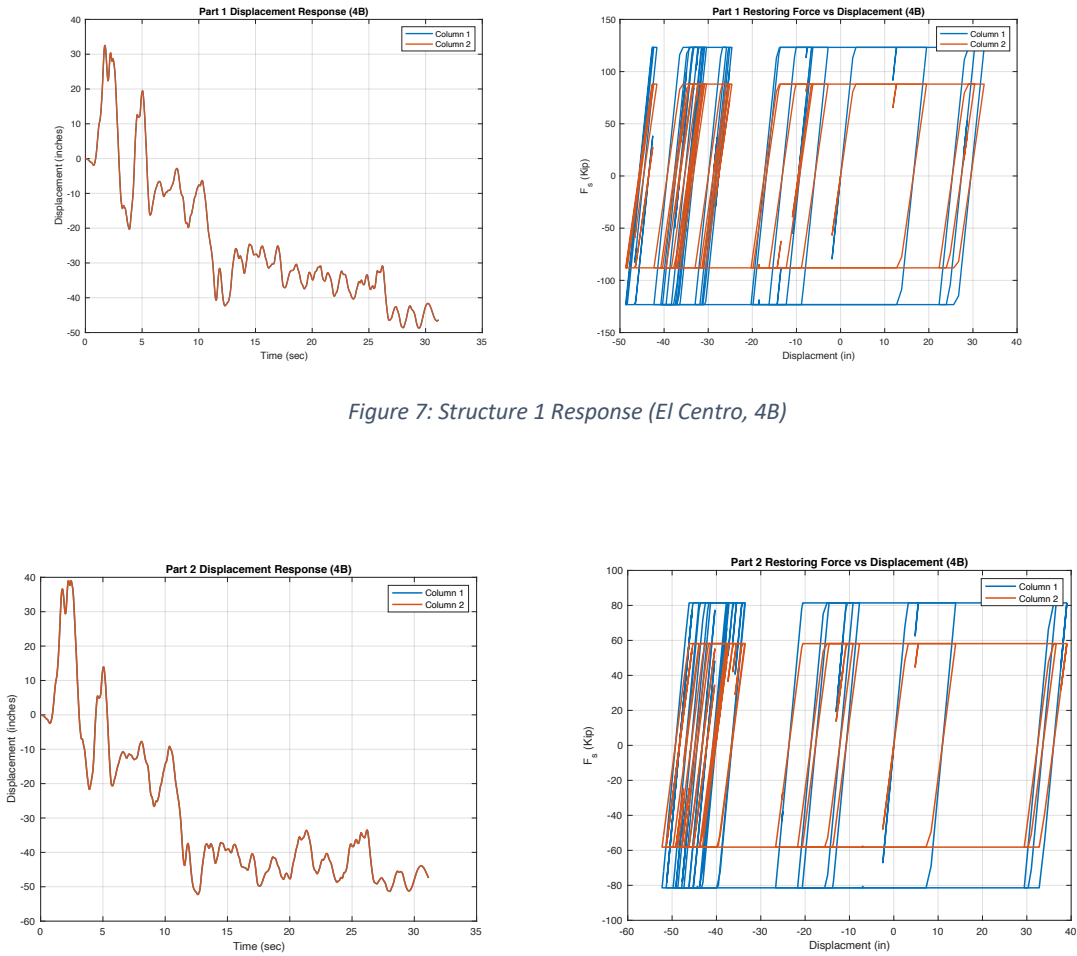


Figure 7: Structure 1 Response (El Centro, 4B)

Figure 8: Structure 2 Response (El Centro, 4B)

PART C

The same procedure shown in part a is used to determine the scaling factor. Part C now uses the Rinaldi 1994 Northridge Earthquake recording.

$$SF = \frac{S_{a_fund}}{S_{a_SDOF}} = 0.7538 \dots (14)$$

The response values of both structures subjected to the Rinaldi 1994 Northridge Earthquake with a scaling factor that corresponds to that of the design spectrum can be seen in table 4.3.

The maximum displacement of structure 1 for both columns is 12.082 inches. Column 1 continues to have a larger restoring force compared to that of column 2 which is expected. Looking at structure 2, the maximum displacement is 11.619 inches for both columns. Where the same relationship between strengths of column in structure 1 is seen in structure 2, with column 1 being stronger. Overall, the displacement decreases from structure 1 to structure 2. This primarily has to do with the two methods discussed earlier, the equal displacement assumption as well as the displacement-based method. The displacement in structure 1 is approximately 1.04 times larger compared to structure 2.

Parameter	Value	Unit
Max Displacement (Structure 1, both col)	12.082	in
Max restoring force (Structure 1,Col1)	123.23	kip
Max restoring force (Structure 1,Col2)	88.018	kip
Max Displacement (Structure 2)	11.619	in
Max restoring force (Structure 2,col1)	81.429	Kip
Max restoring force (Structure 2,col2)	58.164	Kip

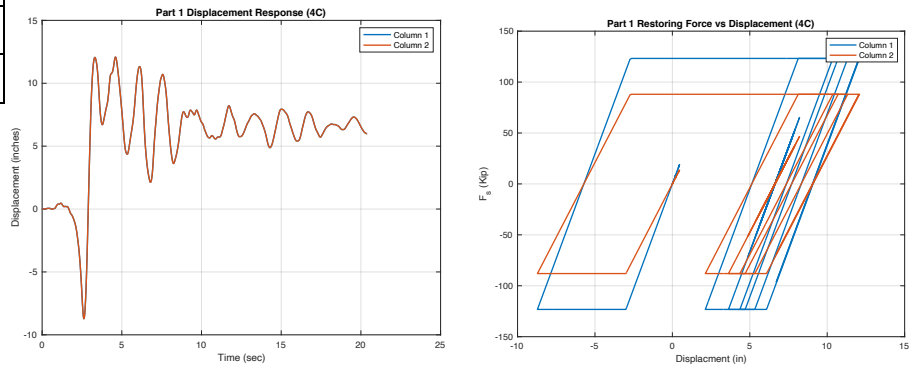


Figure 9: Structure 1 Response (Rinaldi, 4C)

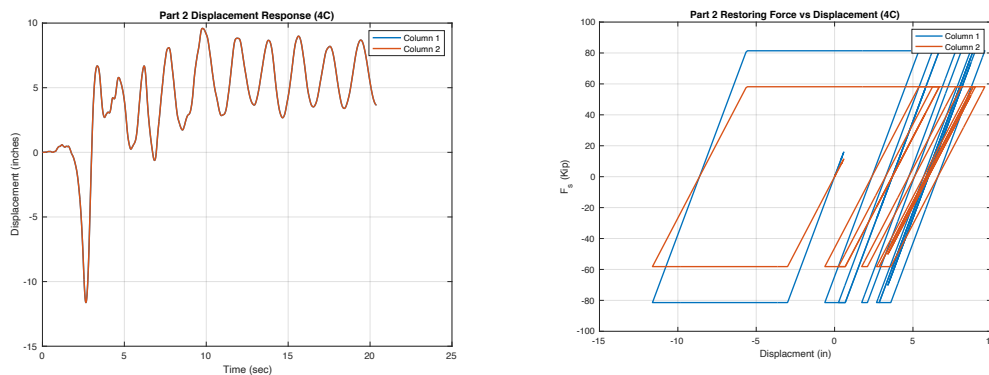


Figure 10: Structure 2 Response (Rinaldi, 4C)

PART D

The same procedure shown in part a is used to determine the scaling factor. Part D now multiplies the fundamental acceleration by a factor of 2.

$$SF = \frac{2S_{a_fund}}{S_{a_SDOF}} = 1.5076 \dots (15)$$

The response values of both structures subjected to the Rinaldi 1994 Northridge Earthquake with a scaling factor that corresponds to that of two times the design spectrum can be seen in table 4.4.

Looking at the maximum values in table 4.4, it can be seen that structure 2 now exhibits the larger displacement rather than structure 1. The displacement in structure 2 is approximately 1.25 times larger than that of structure 1. The restoring force of both structures remain the same compared to that of part C with the same Northridge earthquake. Therefore, it can be seen that the scale factor only affects the displacement. The response of each structure can be seen below.

Table 4.4: Response Values Part D		
Parameter	Value	Unit
Max Displacement (Structure 1, both col)	21.729	in
Max restoring force (Structure 1,Col1)	123.23	kip
Max restoring force (Structure 1,Col2)	88.018	kip
Max Displacement (Structure 2)	27.042	in
Max restoring force (Structure 2,col1)	81.429	Kip
Max restoring force (Structure 2,col2)	58.164	Kip

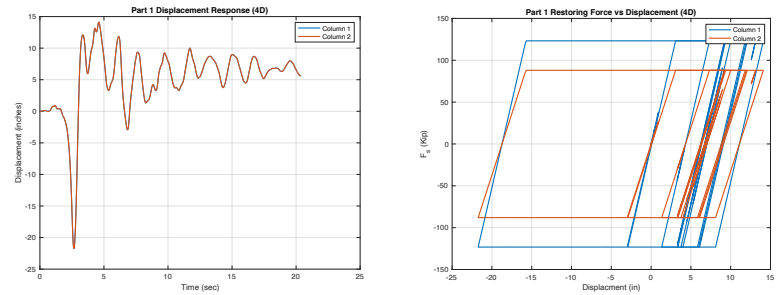


Figure 11: Structure 1 Response (Rinaldi, 4D)

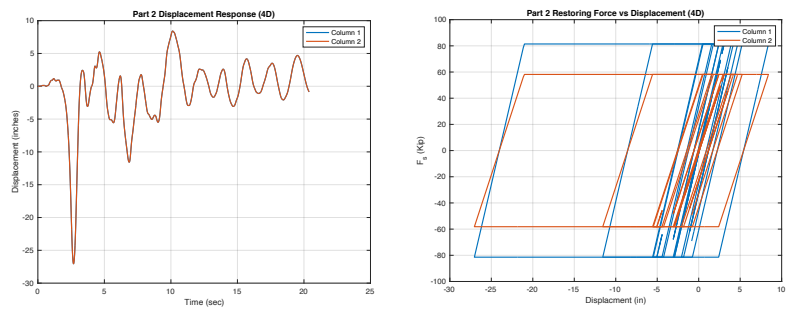


Figure 12: Structure 2 Response (Rinaldi, 4D)

Problem 5

The yield displacement is adjusted from $u_y = 3$ inches to $u_y = 1$ inch. The structure to be analyzed in this section is the structure from problem 2 where the displacement-based method is used along with the equivalent linear elastic assumption. The yield displacements are to change but the yield strengths are required to stay the same. The El Centro recording with a scale factor of $SF = 7.3307$ from part 4b is to be used for analysis. In addition, the Rinaldi Northridge record with a scale factor equal to 1.5076 is to be used for analysis.

The mass matrix will stay the same as the mass matrix shown earlier in the report. Although, the stiffness matrix will now change. The new stiffness of each column can be computed using the relation of $F = KU_y$ where the force (F) is the yield strengths shown in problem 2 and U_y is now adjusted to 1 inch. The new stiffness matrix can be seen below, where the units are kip/in.

$$\widetilde{K}_{prob5} = \begin{bmatrix} 81.429 & 0 \\ 0 & 58.164 \end{bmatrix}$$

The damping matrix can be formulated using the same procedure shown in problem 4 for the Rayleigh damping matrix. The natural frequencies for the modified system in part 5 are: 5.7679 and 6.567 rad/s. Therefore, the damping matrix is

$$\widetilde{C}_{prob5} = \begin{bmatrix} 1.3402 & 0.071582 \\ 0.071582 & 0.93682 \end{bmatrix}$$

The response values of the structure subjected to the El Centro earthquake with a scaling factor of 7.3307 can be seen in table 5.1. The maximum displacement of both columns was the same. Comparing the structure from problem 5 to the structure from problem 4B, with the same recording it is observed that the maximum restoring forces for both are the same although the structure in problem 5 has now experienced a smaller maximum displacement compared to that of 4B. The maximum displacement of both columns is seen to be 1.5 times less than the structure in 4B.

Table 5.1: Response Values Problem 5 (El Centro)		
Parameter	Value	Unit
Max Displacement (Structure, both col)	35.016	in
Max restoring force (Structure,Col1)	81.429	kip
Max restoring force (Structure,Col2)	58.164	kip

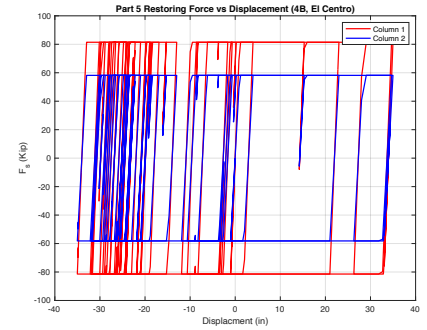
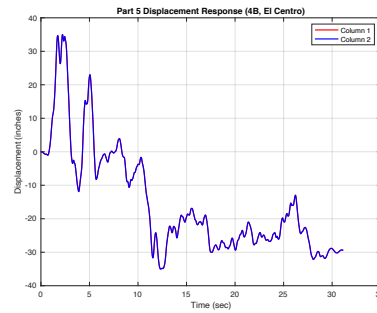


Figure 13: Structure Response (El Centro, Prob 5)

Analysis for the Rinaldi record with a scale factor of 1.5 is applied to the structure in part 5. The response values of the structure can be seen in table 5.2.

Table 5.2: Response Values Problem 5 (Rinaldi)		
Parameter	Value	Unit
Max Displacement (Structure, both col)	20.813	in
Max restoring force (Structure,Col1)	81.429	kip
Max restoring force (Structure,Col2)	58.164	kip

Comparing the response of the structure in problem 5 to that of the structure in problem 4D, shows that the restoring forces remain the same although the displacement of the structure in problem 5 is approximately 1.3 times smaller than that in part 4D. Overall the structure in problem 5 experienced smaller maximum displacements due to the fact that the required stiffness for each column increases by decreasing the yield displacement.

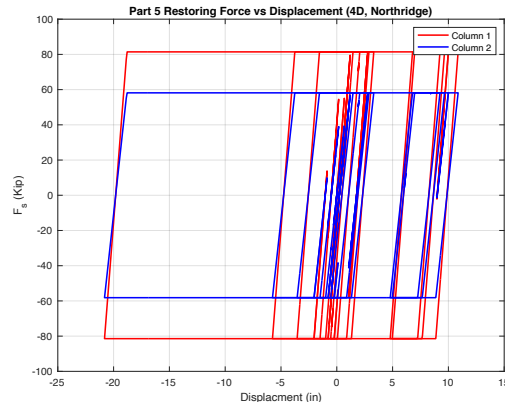
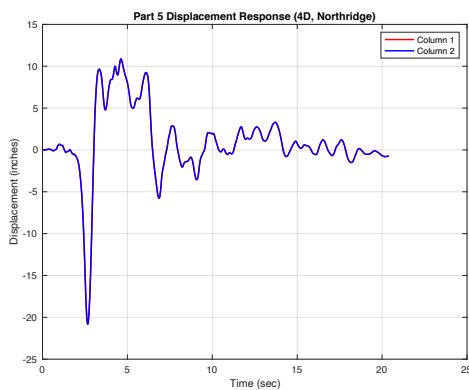


Figure 14: Structure Response (Rinaldi, Prob 5)

Problem 6

In this problem, the structure in problem 6 experienced a yield resistance reduction of 25% therefore the yield strength designed in problem 2 is dropped to 61.072 kips and 43.623 kips for columns 1 and 2 respectively. This creates a new yield displacement of 2.25 inches. The stiffness and the damping matrices are the same as those shown in problem 2.

Table 6.1: Response Values Problem 6 (El Centro)		
Parameter	Value	Unit
Max Displacement (Structure, both col)	52.353	in
Max restoring force (Structure,Col1)	61.072	kip
Max restoring force (Structure,Col2)	43.623	kip

The response of the structure in problem 6 to the El Centro recording can be seen in table 6.1

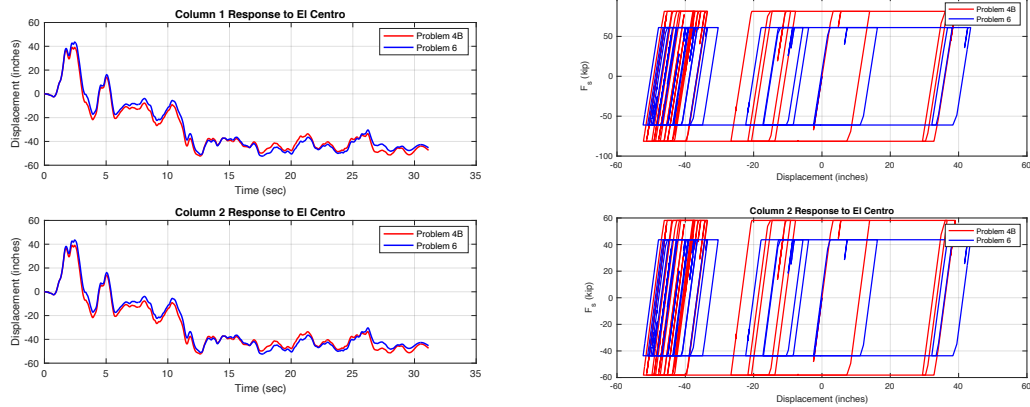


Figure 15: Comparison of 4B and Prob 6 Response (El Centro)

Comparing the response of the structure in problem 6 compared to that in 4B, it is seen that the maximum displacement is nearly identical as there is a 0.17% difference in the values. The restoring forces in the structure in 4B for column 1 and 2 decreased by 25% .

The response of the structure in problem 6 to the Rinaldi recording can be seen in table 6.2. A scale factor of 1.5076 is applied.

Table 6.2: Response Values Problem 6 (Rinaldi)		
Parameter	Value	Unit
Max Displacement (Structure, both col)	28.502	in
Max restoring force (Structure,Col1)	61.072	kip
Max restoring force (Structure,Col2)	43.623	kip

Comparing the structural response in problem 6 to that of the structural response in problem 4D for the Northridge earthquake, it can be seen that the maximum displacements of each column are similar as there approximately 5% differences between the two. A comparison of the structure in problem 4D to that of the structure in 4B can be seen in the figures below as the responses are overlaid.

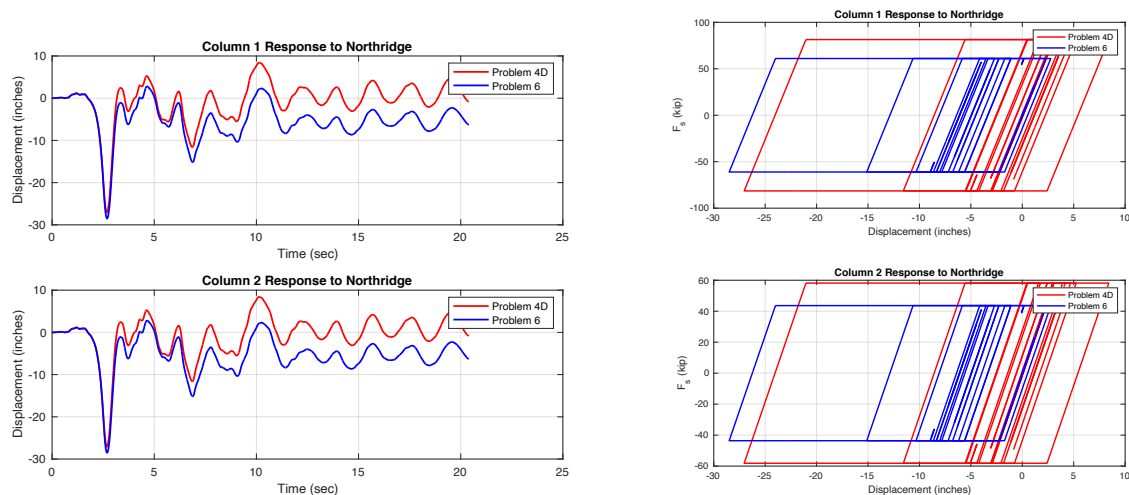


Figure 16: Comparison of 4B and Prob 6 Response (Rinaldi)

Problem 7

In this problem the two columns are to have the same stiffness and yield resistances as the stronger column designed in part 2. The stronger column in part 2 was column 1 (see figure 1 for reference). The stiffness' of the columns are 27.143 kip/in. This will create a 2x2 diagonal K matrix with these entries. The mass matrix will remain the same. In addition, similar to what was shown in problem 2, eigenvalue analysis can be performed to determine the Rayleigh damping matrix. Solving the natural frequencies of this structure, it is seen that the natural frequencies are 3.4465 and 4.3346 rad/s. The damping matrix can be formulated and is shown below.

$$\widetilde{C}_{prob7} = \begin{bmatrix} 0.774 & 0.044755 \\ 0.044755 & 0.63974 \end{bmatrix}$$

The response of the structure in problem 7 to El Centro ground motion can be seen in figure 7.1 An appropriate scale factor of 7.3307 is applied to mimic problem 4B.

Table 7.1: Response Values Problem 7 (El Centro)		
Parameter	Value	Unit
Max Displacement (Structure, col 1)	52.313	in
Max Displacement (Structure, col 2)	51.786	in
Max restoring force (Structure, Col1)	81.429	kip
Max restoring force (Structure, Col2)	81.429	kip

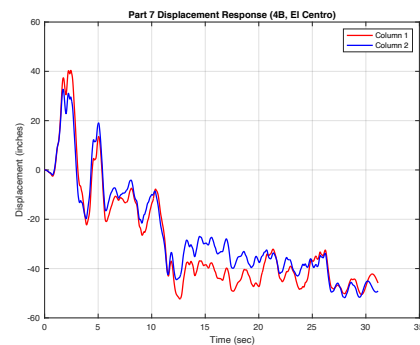
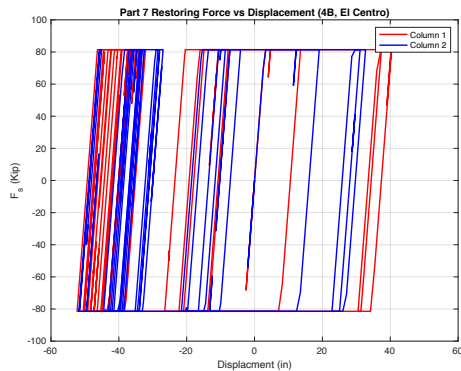


Figure 17: Structure Response (El Centro, Prob 7)



As seen above, both of the columns no longer have the same response. Column 1 experiences a slightly larger displacement compared to column 2. In problem 4B, both columns experienced the same displacement response. The maximum displacement in part 4B is 52.26 inches. These are similar to the values in problem 7, with a 0.1% percent difference for column 1 and a 0.9% difference for column 2. Since the stiffness of each column is now the same, both columns have the same maximum restoring force in problem 7. The restoring force of column 1 is the same as that of problem 2 but it is no observed that the restoring force in column 2 has increased by approximately 40%.

The response of the structure in problem 7 to the Northridge earthquake can be seen in figure 7.2 An appropriate scale factor of 1.5076 is applied to mimic problem 4D.

As see, the columns do not exhibit the same behavior, the stiffness of the columns are the same as a result, the maximum restoring force does not change and remains at 81.429 kips. The structure in problem 4D shows that the maximum restoring force for column 1 is 81.429 kips and 58.164 kips for column 2. This 40% increase in restoring force for column 2 is seen. The maximum displacement observed in problem 4D is 27.042 inches for both columns. This is very similar to column 1 in problem 7. The percent difference observed for maximum displacements between problem 4D and problem 7 are 2.12% for column 1 and 17.5 percent for column 2.

Table 7.2: Response Values Problem 7 (Northridge)		
Parameter	Value	Unit
Max Displacement (Structure, col 1)	27.616	in
Max Displacement (Structure, col 2)	22.299	in
Max restoring force (Structure,Col1)	81.429	kip
Max restoring force (Structure,Col2)	81.429	kip

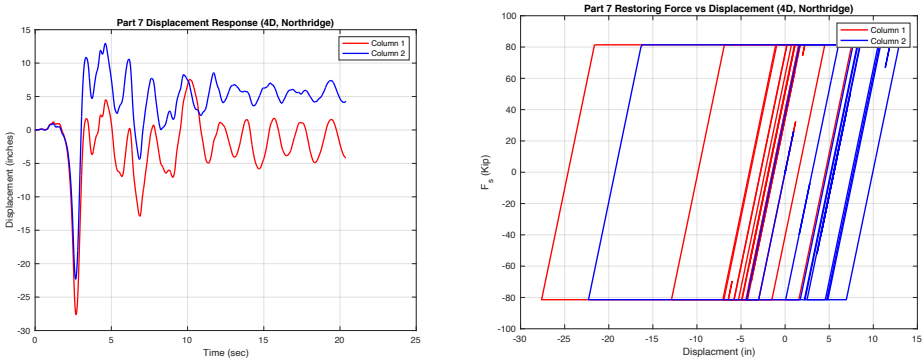


Figure 18: Structure Response (Rinaldi, Prob 7)

Time history analysis of problems 4B and 4D are overlain on problem 7 for comparison.

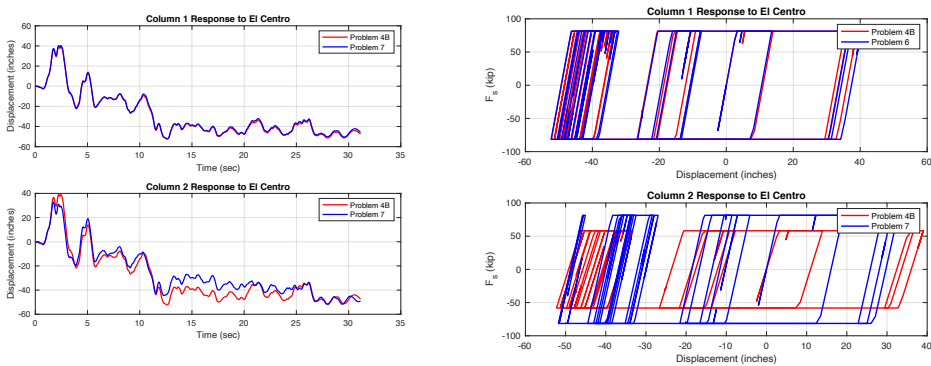


Figure 19: Structure Response Comparison (4B & Prob 7, El Centro)

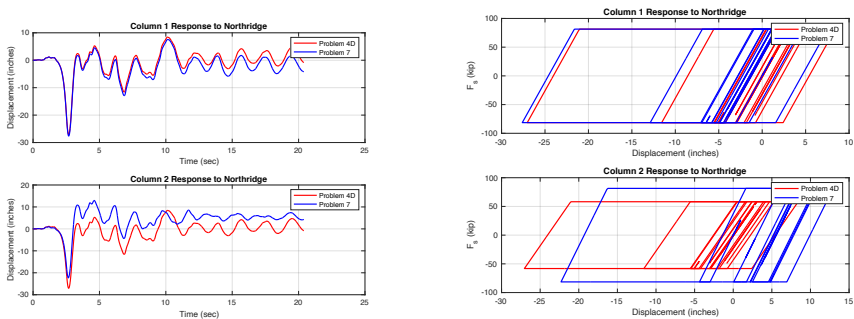


Figure 20: Structure Response Comparison (4D & Prob 7,Rinaldi)

Problem 8: Concluding Remarks

PART A

Force based design along with displacement design provide different results and different responses based on each earthquake ground motion. The forced based design provides a larger required stiffness of the columns compared to the displacement-based design. In addition, when the fundamental period was amplified by two times that of the design spectrum the displacement-based design showed large maximum displacements (structure 2). On the other hand, when the scaling factor was just applied with no amplification (2 times the design spectrum), the displacement-based design displayed smaller maximum displacements for each column and is usually more accurate compared to force-based design. The displacement-based design relies on the stiffness and damping matrices, as seen in problem 2 with the equivalent damping, so any change to these can cause a shift in response.

PART B

Looking at the El Centro and Rinaldi ground motions, it can be seen that the Rinaldi ground motions, in correlation with the fundamental period resulted in smaller scaling factors compared to that of the El Centro ground motion. Therefore, the scaling factors for the Rinaldi recordings were 0.75379 and 1.5076 for parts 4C and 4D compared to the scaling factors of 3.6654 and 7.3307 for the El Centro ground motions. As a result, the Rinaldi methods showed a smaller amplitude response since that specific recording was closely aligned to the fundamental acceleration.

PART C

The elastic period in part 1 (force-based design) is 1.53 seconds while the elastic period in part 2 (displacement-based design) is 1.88 seconds. Knowing the relationship of period (T) to natural frequency, $T = \frac{2\pi}{\omega_n}$, it can be seen that the smaller the period, the stiffer the structure is. A stiffer structure will result in less displacement and smaller ductility as evidenced in problem 4.

PART D

Impact of the yield resistance of a structure on the maximum displacement can be seen in problem 6. The reduction of the yield resistance by 25% resulted in slightly larger displacements and ductility for both the El Centro and the Rinaldi earthquake. Although the values are close, there is an increase that is observed. A comparison before and after the reduction in yield resistance can be seen in the problem 6 section.

PART E

The distance between the center of gravity and where the centroid of rotation is known as the eccentricity and the mass acting an eccentric distance is known as the eccentric mass. If the center of mass is close to the stiffness centroid, small rotations will occur and thus less displacements. If the mass eccentricity is close to the stiffness centroid, smaller ductility values will be seen. The 3-frame structure in this analysis is not symmetric, therefore significant rotation would have occurred if the columns were not properly designed, which is why the tributary lengths and mass were taken into account in problems 1 and 2.

PART F

The main observation seen throughout this analysis is the importance of scaling factors. Scaling factors are used to get the ground recording response to be as close as possible to the design spectrum in ASCE 7, thus leading in a more accurate analysis. While studying the concept of scaling factors it was seen that ASCE 7 puts a limit for amplitude scaling. According to ASCE 16.2.3.2 “The average of the maximum-direction spectra from all the ground motions shall not fall below 90% of the target response spectrum for any period within the same period range”.

```

function [r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug)
%MODIFIED NEWTON RAPHSON FOR CONSTANT AVERGAGE ACC METHOD
%Set the tolerance for the Newton Phaphson (Modified) method✓
according to
%the textbook ch 5, requirement 1
tolerence = 10^(-5)*ones(dof,1);

%Forcing applied from the Earthquake, NEWTONS law  $F = m*acc$ ,✓
where a in the
%equation is teh driection of applied loading.
    forcing = -M*a*ddug; %F = MA

%Const Avg ACC Parameters, given in Dr. Shings notes
beta = 0.25; %CONST AVG ACC METHOD
gamma = 0.5; %CONST AVG ACC METHOD

% NEWTON RAPHSON WITH CONST AVERAGE ACC METHOD from lecture 19
    %Slides 11-14 with steps 3 in the problem statement✓
(material law)

%STEP 1 create empty vectors to store values
u = zeros(dof,length(t));✓
%Initialize DISP
du = zeros(dof,length(t));✓
%Initialize VEL
ddu = zeros(dof,length(t));✓
%Initialize ACC
%IN Dr. Shings Notes these are not the dispalcmeents to be✓
ploted
u_bar = zeros(dof,length(t));✓
%Initialize DISP BAR
du_bar = zeros(dof,length(t));✓
%Initialize VEL BAR
ddu_bar = zeros(dof,length(t));✓
%Initialize ACC BAR
K_bar = zeros(dof,dof,length(t));
r = zeros(dof,length(t));
deltau_trial = zeros(dof,length(t));
increment
u_t = zeros(dof,length(t));
du_t = zeros(dof,length(t));

```

```

% "Fs" Restoring Force
%Trial change in time✓
%Trial displacement
%Trial Velocity

```



```

ddu_t = zeros(dof,length(t));           %Trial Acceleration
r_try = zeros(dof,length(t));           %trial restroing force

```

%NOTE: ALL trial values are inside while loop, essentially do✓
not need to
%be stored buy in this case we are.

```

%Time increment Loop, first initiation
for i = 1:(length(t)-1)

```

```

    %STEP 2 from Notes
    u_bar(:,i+1) = u(:,i) + dt*du(:,i) + (dt^2)*(0.5 - beta)*ddu✓
(:,i); %SLIDE 11
    du_bar(:,i+1) = du(:,i) + dt*(1-gamma)*ddu(:,i);✓
%SLIDE 11
    % STEP 3 from Notes
    u_t(:,i+1) = u_bar(:,i+1);✓
%SLIDE 11
    du_t(:,i+1) = du_bar(:,i+1);✓
%SLIDE 11
    ddu_t(:,i+1) = zeros(dof,1);✓
%SLIDE 11
    delta_u = u_t(:,i+1) - u(:,i+1);           %✓
Comes from problem statment (material Law) %SLIDE 11
    r_trial = K*u_t(:,i+1);                     %✓
Comes from problem statment (material Law) %SLIDE 11
    r_try(:,i+1) = sign(r_trial).*min(abs(r_trial),ry'); %✓
Comes from problem statment (material Law) %SLIDE 11
    % STEP 4
    K_bar(:, :, i+1) = (1/(dt^2*beta))*M + (gamma/(dt*beta))*C +✓
K; %SLIDE 12
    % STEP 5
    Res(:,i+1) = forcing(:,i+1) - M*ddu_t(:,i+1) - C*du_t(:,i+1)✓
- r_try(:,i+1); %SLIDE 12
    % STEP 6
    deltau_trial = inv(K_bar(:, :, i+1))*Res(:,i+1); %✓
Powerpoint SLIDE 12
    % STEP 7
    u_t(:,i+1) = u_t(:,i+1) + deltau_trial; % Powerpoint✓
SLIDE 12, update trial soln
    % STEP 8

```



```

    delta_u = u_t(:,i+1) - u(:,i); %Comes ✓
from problem statment (material Law) %SLIDE 12
    r_trial = r_try(:,i) + K*delta_u; %Comes ✓
from problem statment (material Law) %SLIDE 12
    r_try(:,i+1) = sign(r_trial).*min(abs(r_trial),ry'); %Comes ✓
from problem statment (material Law) %SLIDE 12
    % STEP 9
    ddu_t(:,i+1) = (1/(dt^2*beta))*(u_t(:,i+1) - u_bar(:,i+1)); ✓
%Powerpoint SLIDE 13
    du_t(:,i+1) = du_bar(:,i+1) + dt*gamma*ddu_t(:,i+1); ✓
%Powerpoint SLIDE 13

    % IMPLEMENT WHILE LOOP FOR TOLERANCE %Powerpoint SLIDE 12,
    while abs(Res(:,i+1)) > tolerance %REsidual ✓
requirement %Powerpoint SLIDE 12, step 10
    %STEP 4 AGAIN, essentially we repeat steps 4-12 on slides ✓
12-13 of
    %lecture 19 until we exit the while loop
    K_bar(:, :, i+1) = (1/(dt^2*beta))*M + (gamma/(dt*beta))*C + ✓
K; %K will stay the same MODIFIED NEWTON
    % STEP 5 AGAIN (K loop)
    Res(:,i+1) = forcing(:,i+1) - M*ddu_t(:,i+1) - C*du_t(:,i+1) ✓
- r_try(:,i+1);
    % STEP 6 AGAIN (K loop)
    deltau_trial = K_bar(:, :, i+1)\Res(:,i+1);
    % STEP 7 AGAIN (K loop)
    u_t(:,i+1) = u_t(:,i+1) + deltau_trial;
    % STEP 8 AGAIN (K loop)
    delta_u = u_t(:,i+1) - u(:,i);
    r_trial = r_try(:,i) + K*delta_u;
    r_try(:,i+1) = sign(r_trial).*min(abs(r_trial),ry');
    %STEP 9 AGAIN (K loop)
    ddu_t(:,i+1) = (1/(dt^2*beta))*(u_t(:,i+1) - u_bar(:,i+1));
    du_t(:,i+1) = du_bar(:,i+1) + (dt*gamma)*ddu_t(:,i+1);
end
    % Step 12 STORE VALUES FOR PLOTS (Powerpoint Slide 13)
    r(:,i+1) = r_try(:,i+1); %Residual Force essentially FS ✓
used to obtain plots
    u(:,i+1) = u_t(:,i+1); %dispalcmeent
    du(:,i+1) = du_t(:,i+1); %velocity, not really needed or ✓
used

```

```
ddu(:,i+1) = ddu_t(:,i+1); %Acceleration
```

```
end
```

```
end
```

```
function [u, du, ddu] = LinearTH2(M,C,K,dt,Time,Dof,a,Acc)
```

```
%SAME Code from HW 5 , used for SDOF analysis to determine the✓  
displacement  
%Sd
```

```
gamma = 0.5; %value for constant-average-acceleration method  
beta = 0.25; %value constant-average-acceleration method
```

```
u = zeros(length(Time),Dof);  
du = zeros(length(Time),Dof);  
ddu = zeros(length(Time),Dof);  
f = zeros(length(Time),Dof);  
Forcing = zeros(length(Time),Dof);
```

```
%initiate the For loop
```

```
for j = 1:length(Time) %Just incase we have more time steps we want to✓  
check make a dt vector
```

```
%Step 2 (Determine the effective stiffness)
```

```
K_eff = (1/(beta*dt(j)^2))*M + (gamma/(beta*dt(j)))*C + K;
```

```
%THE FOLLOWING STEPS WERE TAKEN FROM SLIDE 5 OF LECTURE✓  
"TIME INTEGRATION  
%METHODS"
```

```
for i = 1:(length(Time)-1)
```

```
    t(i+1) = i*dt;
```

```
%UPDATE TIME
```

```
    f = M * a * [Acc(i+1)*386.4];
```

```
    Forcing(i,:) = f;
```

```
%Step 3: Compute the effective force
```

```
%f_i+1 comes from the El Centro Recording, which is the✓  
forcing
```

```
%applied
```

```
    f_eff(i+1,:) = f + ((1/(beta*dt^2))*M + (gamma/(beta*dt)✓  
*C))*u(i,:)' + (((1/(beta*dt))*M + ((gamma/beta)-1)*C)*du(i,:)' )✓  
+ (((1/(2*beta)-1)*M + dt/2*((gamma/beta)-2)*C))*ddu(i,:)' ;
```

%Step 4: Solve for the displacement at u_{i+1}

$u(i+1,:) = f_eff(i+1,:)/K_eff;$

%Step 5: Compute the acceleration at ddu_{i+1}

$ddu(i+1,:) = (1/(beta*dt^2))*(u(i+1,:) - u(i,:) - dt*du(i,:) - (1/2 - beta)*dt^2*ddu(i,:));$

%Step 6: Compute the velocity at ddu_{i+1}

$du(i+1,:) = du(i,:) + dt*((1-gamma)*ddu(i,:) + gamma*ddu(i+1,:));$
 end

end
 end

```

%% TERM PROJECT, SE 221, ALLEND OSMAN BOTANI
% DR SHING
%TA: LUCAS DEYGLUN
clear all; close all; clc;
format shortg
%%LOAD EL CENTRO DATA%%
g = 386.09; %Gravity in✓
in/s^2
Data = load('El_Centro_EQ-Motion.txt');
Time = Data(:,1);
Acc = Data(:,2).*g; %in/s^2

%%LOAD NORTHRIDGE DATA from CANVAS%%
Data = importdata('Northridge-Rinaldi.txt');
Rinaldi = Data.data*g; %in/s^2
Rinaldi = Rinaldi(1:10196);
dt2 = 0.002; %new dt not same✓
s El centro
Time2max = length(Rinaldi)*dt2;
Time2 = linspace(0,Time2max,length(Rinaldi));

%%Given Parameters%%
DOF =2;
L1 = 40; L2 = 120; L3 = 20; %ft
Lgir = 180; %length of girder ft
Ltrib = [Lgir/2 - L3;Lgir/2 - L1]; %trib length in "ft"
rtrib = Ltrib/L2; %trib mass
w = 9; %Girder weight, kip/ft
wtotal = w*Lgir; %total weight of girder✓
in kips
M = wtotal/g; %mass per unit length✓
k*s^2/in
xi = 0.05; %Damping (same for bridge✓
and columns)

%Response spectrum values(Bridge design)
S_ds = 1.5*g; %units in in/s^2
S_d1 = g; %units in in/s^2
T_s = S_d1/S_ds; %Seconds
T_l = 12; %Seconds

```

%Given values for the displacment in problem statement

uy = 3; %yield dsiaplcement in inches

umax = 15; %Max dsiaplcement in inches

%% PART 1 %%

clc;

mu = umax/uy; %max permitted ductility demand

%Using Relations for each column

disp('Part 1')

dt = 1e-6; %Time increment (Arbitrary)

Tend = 14; %end point for period vector

T = [0:dt:Tend]';

n = size(T);

%Create response spectrum ACC (Figure 2) , a peicewise✓
function/plot

Sa = zeros(n); %initialize ACc

for i = 1:n

if T(i) <=T_s

Sa(i) = S_ds;

elseif T(i) <= T_l

Sa(i) = S_d1/T(i);

else T(i) > T_l;

Sa(i) = S_d1*T_l/(T(i)^2);

end

end

V = wtotal*Sa/(g*mu); %Shear (max)

Kreq = V/uy; %min stiff vector to satisfy disp

%determine teh elastic period by trial and error

Trialperiod = zeros(length(T),1);

dt= zeros(length(T),1); %Trial Timestep

%Elastic period

for i = 1:length(T)

Trialperiod(i) = 2*pi/(sqrt(Kreq(i))/(M)); % T = 2pi/wn

```

    dtt(i) = abs(T(i) - Trialperiod(i));           %delta t
end
Te_1 = T(find(min(dtt) == dtt));                 %Search for where✓
the index is
fprintf('Telastic = %.3f sec\n',Te_1)

%Stiffness and yield strength via iteration
K_eq = Kreq(find(min(dtt) == dtt));              %Search for where✓
the index is in the minimum stiff
for i = 1:DOF
    ry(i) = rtrib(i) * K_eq * uy;                %Strength, yield✓
    relationship (girder is not symmeytric)
    K(i) = ry(i)/uy;                             % Stiffness based✓
on F = KU
end
K_p1 = K;
ry_p1 = ry';
Keq_p1 = K_eq';
fprintf('K1 = %.3f kip/in\n',K(1))
fprintf('K2 = %.3f kip/in\n',K(2))
fprintf('ry1 = %.3f kips\n',ry(1))
fprintf('ry2 = %.3f kips\n',ry(2))
%% PART 2
disp(' ') %Just a divider for the output
disp('PART 2')
DOF = 2;
ti = 0.1;
K_eq1 = ry(1)/umax; % Equivalent Stiffness of one column using✓
relation of ry = K_eq*Umax
EHD1 = (2*ry(1))*2*(umax-uy); %kip-in, area under curve of✓
equivalent linear cycle
xi_eq1 = EHD1/(2*pi*umax^2*K_eq1); %Equivalent damping
K_eq2 = ry(2)/umax; % Equivalent Stiffness of one column using✓
relation of ry = K_eq*Umax
EHD2 = (2*ry(2))*2*(umax-uy); %kip-in, area under curve of✓
equivalent linear cycle
xi_eq2 = EHD2/(2*pi*umax^2*K_eq2); %Equivalent damping
xi_eq = xi_eq2; %Same for both cases
fprintf('xi_eq = %.3f \n',xi_eq)

%Eurocode Amplification

```

```

term = sqrt(0.07/(0.02+xi_eq));

%PSEUDO ACC PLOT
figure
plot(T,Sa,'linewidth',1.5)
title('Design Spectrum: ACC')
xlabel('Period (Sec)')
ylabel('S_A (in/s^2)')
hold on
xline(T_s,'--','T_s');
hold on
xline(T_l,'--','T_l');
grid on
hold off

%Displacmenet Repsonse Spectrum
for i = 1:length(T)
    Sd(i) = Sa(i)*(T(i)/(2*pi))^2; %unit conversion in/s^2 --✓
> in.
    Sd_eq(i) = Sd(i) * term;
end

%Plot DISP Response Spectrum
figure
plot(T,Sd_eq,'r','linewidth',1.75)
hold on
plot(T,Sd,'g','linewidth',1.75)
hold on
yline(umax,'--','U_m_a_x');
title('Design Spectrum: Displacement')
xlabel('Period (Sec)')
ylabel('S_d (in)')
legend("\xi_e_q ", "\xi")
xlim([0 5])
grid on

%Determine where the equalivent period is
for i = 1:length(T)
    if round(Sd_eq(i),2) == 15.0
        Tlocation = i;
    end
end

```


end

%Determine Teq and Keq and Design Base Shear (Equivalent Values)

```
T_eq_part2 = T(Tlocation);
K_eq_part2 = (4*pi^2*M/(T_eq_part2^2));
V_eq_part2 = K_eq_part2*umax;
K_eq_yield= V_eq_part2/uy;
Te_2 = 2*pi/sqrt(K_eq_yield/M);
fprintf('T_eq = %.3f sec \n',T_eq_part2)
fprintf('K_eq = %.3f kip/in \n',K_eq_part2)
fprintf('K_eq_yield = %.3f kip/in \n',K_eq_yield)
fprintf('V_eq = %.3f kip \n',V_eq_part2)
fprintf('T_elastic = %.3f sec \n',Te_2)
```

%Using Relations for each column

```
K2_part2 = V_eq_part2/(uy*(70/50 + 1)); %From statics
(Relationship between col 1 and 2 w moments at pin)
K1_part2 = 1.4*K2_part2;
ry1_part2 = K1_part2*uy; %kip,yield strength
for first column
ry2_part2 = K2_part2*uy; %kip,yield strength
for first column
wn_part2 = [sqrt(K1_part2/M) sqrt(K2_part2/M)]';
ry_p2 = [ry1_part2;ry2_part2];
fprintf('K1_part2 = %.3f kip/in \n',K1_part2)
fprintf('K2_part2 = %.3f kip/in \n',K2_part2)
fprintf('ry1_part2 = %.3f kip \n',ry1_part2)
fprintf('ry2_part2 = %.3f kip \n',ry2_part2)
```

%% PART 3 JUST THE CODE ATTACHED, NO CALCULATIONS RAN

```
close all;
disp(' ') %Just a divider for the output
disp('PART 3')
```

%Establish the Rayleigh Damping Matrix

```
BigM = zeros(DOF,DOF,2);
BigM(:,:,1) = M*[19/36 1/18; 1/18 13/36];
BigM(:,:,2) = BigM(:,:,1);
```

```
BigK = zeros(DOF,DOF,2);
```

```

BigK(:,:,2) = [K1_part2    0
               0    K2_part2];
BigK(:,:,1) = [K_p1(1)    0
               0    K_p1(2)]; %Stiff Matrix

BigC = zeros(DOF,DOF,2);
%Solve for natural frequencies
[V1,D] = eig(BigK(:,:,1),BigM(:,:,1),"vector");
wn_p1 = sqrt(D);
wn1 = wn_p1(1); wn2 = wn_p1(2);
%SOLVE SYTEM TO GET a0 and a1
wnmat = [1/wn1 wn1;
         1/wn2 wn2];
avals = 2* inv(wnmat)*[xi;xi];
a0 = avals(1);
a1 = avals(2);
BigC(:,:,1) = a0*BigM(:,:,1) + a1*BigK(:,:,1);

%Repeat for Part2 Matricies
[V2,D2] = eig(BigK(:,:,2),BigM(:,:,2),"vector");
wn_p2 = sqrt(D);
wn1 = wn_p2(1); wn2 = wn_p2(2);
%SOLVE SYTEM TO GET a0 and a1
wnmat = [1/wn1 wn1;
         1/wn2 wn2];
avals = 2* inv(wnmat)*[xi;xi];
a0 = avals(1);
a1 = avals(2);
BigC(:,:,2) = a0*BigM(:,:,2) + a1*BigK(:,:,2);

Bigry = zeros(DOF,1,DOF);
Bigry(:,:,1) = ry_p1;
Bigry(:,:,2) = ry_p2;
%% PART 4
clc;
a = [1;1]; %Direction of Applied Loading

%%SECTION A%%
T_fund = (Te_1 + Te_2)/2; %Average of two structures
dt_elcentro = 0.02; %Given time recording increment

```

%Scale Factor Determination (SDOF Assumption)

```
wn_SF = 2*pi/T_fund;  
m_SF = 1; k_SF = m_SF*(wn_SF)^2; %Arbitrary mass value  
c_SF = 2*m_SF*xi*wn_SF;  
    %Run code to get disp SD  
    a1 = [1];  
    [u, du, ddu] = LinearTH2(m_SF,c_SF,k_SF,dt_elcentro,Time,1,✓  
a1,Acc);  
    Sd_ec = max(abs(u))/g;  
    Sa_ec = wn_SF^2 * Sd_ec;  
    Sa_fund = S_d1/T_fund;  
    SF1 = Sa_fund/Sa_ec;
```

%Determine where the fundamental period is

```
for i = 1:length(T)  
    if round(T(i),2) == 1.71  
        Tlocation = i;  
    end  
end
```

%NEWTON RAPHSON

```
dt_recording = 0.02;
```

%%PART 1%%

```
M = BigM(:, :, 1);  
K = BigK(:, :, 1);  
C = BigC(:, :, 1);  
ry = Bigry(:, :, 1)';  
ddug = SF1*Acc';  
dof = DOF;  
t = Time;  
dt = dt_recording;  
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);  
umax4a1_struct1 = max(abs(u(1,:)));  
umax4a2_struct1 = max(abs(u(2,:)));  
rmax4a1_struct1 = max(abs(r(1,:)));  
rmax4a2_struct1 = max(abs(r(2,:)));  
    %Determine where the fundamental period is  
    for i = 1:length(T)
```

```

        if round(T(i),2) == 1.71
            Tlocation = i;
        end
    end
end

```

```

%%Plot Figure%%
figure %DISPLACEMENT
plot(Time,u(1,:), 'linewidth',1.5)
hold on
plot(Time,u(2,:), 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 1 Displacement Response (4A)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 1 Restoring Force vs Displacement (4A)')
hold off
grid on

```

```

%PART 2 (4A)%
%%PART 2%%

```

```

M = BigM(:,:,2);
K = BigK(:,:,2);
C = BigC(:,:,2);
ry = Bigry(:,:,2)';
ddug = SF1*Acc';
dof = DOF;
t = Time;
dt = dt_recording;
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);

```

```
umax4a1_struct2 = max(abs(u(1,:)));
umax4a2_struct2 = max(abs(u(2,:)));
rmax4a1_struct2 = max(abs(r(1,:)));
rmax4a2_struct2 = max(abs(r(2,:)));
    %Determine where the fundamental period is
    for i = 1:length(T)
        if round(T(i),2) == 1.71
            Tlocation = i;
        end
    end

    unscale_acc = Acc';
    wn_fund = (2*pi)/ T_fund;
    sd_fund = Sd(Tlocation);
    sa_fund = Sa(Tlocation);
    sa_pseudo = wn_fund^2 * sd_fund;

%%Plot Figure%%
figure %DISPLACEMENT
plot(Time,u(1,:), 'linewidth',1.5)
hold on
plot(Time,u(2,:), 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 2 Displacement Response (4A)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 2 Restoring Force vs Displacement (4A)')
hold off
grid on
```

```
%% Section B%% BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB
```

```
%%PART 1%%
```

```
M = BigM(:, :, 1);
K = BigK(:, :, 1);
C = BigC(:, :, 1);
ry = Bigry(:, :, 1)';
ddug = Acc';
dof = DOF;
t = Time;
dt = dt_recording;
```

```
%Run code to get disp SD
```

```
a1 = [1];
[u, du, ddu] = LinearTH2(m_SF, c_SF, k_SF, dt_elcentro, Time, 1, ✓
a1, Acc);
Sd_ec = max(abs(u))/g;
Sa_ec = wn_SF^2 * Sd_ec;
Sa_fund = 2 * S_d1/T_fund;
SF2 = Sa_fund/Sa_ec;
```

```
ddug = SF2* Acc'; %With Scale Factor
[r, u, du, ddu] = PART3_NR(M, C, K, ry, t, a, dof, dt, ddug);
```

```
umax4b1_struct1 = max(abs(u(1, :)));
umax4b2_struct1 = max(abs(u(2, :)));
rmax4b1_struct1 = max(abs(r(1, :)));
rmax4b2_struct1 = max(abs(r(2, :)));
```

```
%Determine where the fundamental period is
```

```
%%Plot Figure%%
```

```
figure %DISPLACEMENT
plot(Time, u(1, :), 'linewidth', 1.5)
hold on
plot(Time, u(2, :), 'linewidth', 1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 1 Displacement Response (4B)')
```

```

hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacment (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 1 Restoring Force vs Displacement (4B)')
hold off
grid on

%PART 2 (4B)%
%%PART 2%%
M = BigM(:,:,2);
K = BigK(:,:,2);
C = BigC(:,:,2);
ry = Bigry(:,:,2)';
ddug = SF2*Acc';
dof = DOF;
t = Time;
dt = dt_recording;
%Newton raphson
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);

umax4b1_struct2 = max(abs(u(1,:)));
umax4b2_struct2 = max(abs(u(2,:)));
rmax4b1_struct2 = max(abs(r(1,:)));
rmax4b2_struct2 = max(abs(r(2,:)));

fourub1 = u(1,:);
fourub2 = u(2,:);
fourrb1 = r(1,:);
fourrb2 = r(2,:);
%Determine where the fundamental period is
for i = 1:length(T)
    if round(T(i),2) == 1.71
        Tlocation = i;
    end
end
end

```

[illegible]


```
%Run code to get SF
```

```
a1 = [1];
[u, du, ddu] = LinearTH2(m_SF,c_SF,k_SF,dt2,Time,1,a1,ddug);
Sd_ec = max(abs(u))/g;
Sa_ec = wn_SF^2 * Sd_ec;
Sa_fund = S_d1/T_fund;
SF3 = Sa_fund/Sa_ec;
```

```
%Newton Raphson
```

```
ddug = SF3*Rinaldi';
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
```

```
umax4c1_struct1 = max(abs(u(1,:)));
umax4c2_struct1 = max(abs(u(2,:)));
rmax4c1_struct1 = max(abs(r(1,:)));
rmax4c2_struct1 = max(abs(r(2,:)));
```

```
%%Plot Figure%%
```

```
figure %DISPLACEMENT
plot(Time2',u(1,:), 'linewidth',1.5)
hold on
plot(Time2',u(2,:), 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 1 Displacement Response (4C)')
hold off
grid on
```

```
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 1 Restoring Force vs Displacement (4C)')
hold off
grid on
```

```
%PART 2 (4C)%
```

```
%%PART 2%%
```

```
M = BigM(:, :, 2);
K = BigK(:, :, 2);
C = BigC(:, :, 2);
ry = Bigry(:, :, 2)';
ddug = SF3*Rinaldi';
dof = DOF;
t = Time2;
dt = dt2;
```

```
%Newton Raphson
```

```
ddug = SF3*Rinaldi';
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
```

```
umax4c1_struct2 = max(abs(u(1,:)));
umax4c2_struct2 = max(abs(u(2,:)));
rmax4c1_struct2 = max(abs(r(1,:)));
rmax4c2_struct2 = max(abs(r(2,:)));
```

```
%%Plot Figure%%
```

```
figure %DISPLACEMENT
plot(Time2,u(1,:), 'linewidth',1.5)
hold on
plot(Time2,u(2,:), 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 2 Displacement Response (4C)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 2 Restoring Force vs Displacement (4C)')
```

```
hold off
grid on
%%
%%Section D%% RINALDI RECORD

%%D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-✓
D-D-D-D-D-D-D-D-D-D-D-D
    %%PART 1%%
M = BigM(:,:,1);
K = BigK(:,:,1);
C = BigC(:,:,1);
ry = Bigry(:,:,1)';
ddug = Rinaldi';
dof = DOF;
t = Time2';
dt = dt2;

%Run code to get SF
a1 = [1];
[u, du, ddu] = LinearTH2(m_SF,c_SF,k_SF,dt2,Time,1,a1,ddug);
Sd_ec = max(abs(u))/g;
Sa_ec = wn_SF^2 * Sd_ec;
Sa_fund = 2*S_d1/T_fund;
SF4 = Sa_fund/Sa_ec;

ddug = SF4 * Rinaldi'; %With Scale Factor

%Newton Raphson
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax4d1_struct1 = max(abs(u(1,:)));
umax4d2_struct1 = max(abs(u(2,:)));
rmax4d1_struct1 = max(abs(r(1,:)));
rmax4d2_struct1 = max(abs(r(2,:)));

%%Plot Figure%%
figure %DISPLACEMENT
plot(Time2',u(1,:), 'linewidth',1.5)
hold on
plot(Time2',u(2,:), 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
```

```

legend('Column 1', 'Column 2')
title('Part 1 Displacement Response (4D)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacment (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 1 Restoring Force vs Displacement (4D)')
hold off
grid on

```

```
%PART 2 (4D)%
```

```
%%PART 2%%
```

```

M = BigM(:, :, 2);
K = BigK(:, :, 2);
C = BigC(:, :, 2);
ry = Bigry(:, :, 2)';
ddug = SF4*Rinaldi';
dof = DOF;
t = Time2;
dt = dt2;

```

```
%NR Computation
```

```

[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax4d1_struct2 = max(abs(u(1,:)));
umax4d2_struct2 = max(abs(u(2,:)));
rmax4d1_struct2 = max(abs(r(1,:)));
rmax4d2_struct2 = max(abs(r(2,:)));

```

```

fourud1 = u(1,:);
fourud2 = u(2,:);
fourrd1 = r(1,:);
fourrd2 = r(2,:);
figure %DISPLACEMENT
plot(Time2,u(1,:), 'linewidth',1.5)
hold on
plot(Time2,u(2,:), 'linewidth',1.5)
xlabel('Time (sec)')

```

```

ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 2 Displacement Response (4D)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 2 Restoring Force vs Displacement (4D)')
hold off
grid on

```

```
SF = [SF1; SF2; SF3; SF4];
```

```

%% Part 6
%% PART 6
clc;close all;
M = wtotal/g;
%Using Relations for each column
K2_part2 = V_eq_part2/(uy*(70/50 + 1));
K1_part2 = 1.4*K2_part2;
ry1_part2 = K1_part2*uy; %kip,yield strength✓
for first column
ry2_part2 = K2_part2*uy; %kip,yield strength✓
for first column
wn_part2 = [sqrt(K1_part2/M) sqrt(K2_part2/M)]';
ry_p2 = [ry1_part2;ry2_part2];

```

```
%Reduce ry by 25%
```

```

ry1_part6 = ry1_part2 * 0.75;
ry2_part6 = ry2_part2 * 0.75;
uy1_p6 = ry1_part6 / K1_part2;
uy2_p6 = ry2_part6 / K2_part2;

```

```

fprintf('uy = %.3f kip/in \n',uy1_p6 )
fprintf('uy = %.3f kip/in \n',uy1_p6 )
fprintf('ry1_part6 = %.3f kip \n',ry1_part6)

```

```

fprintf('ry2_part6 = %.3f kip \n',ry2_part6)
%%PART 1 EL CENTRO %%
M = BigM(:,:,1);
BigK(:,:,1) = [K1_part2      0
               0      K2_part2];
K = BigK(:,:,1);
C = [0.79281 0.05084;0.05084 0.55176]; %NOTE DAMPING MATRIX✓
STAYS THE SAME SINCE K IS THE SAME

Bigry(:,:,1) = [ry1_part6 ry2_part6];
ry = Bigry(:,:,1)';
ddug = SF2*Acc';
dof = DOF;
t = Time;
dt = 0.02; %Given✓
time step in the recording file
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax6b1 = max(abs(u(1,:)));
umax6b2 = max(abs(u(2,:)));
rmax6b1 = max(abs(r(1,:)));
rmax6b2 = max(abs(r(2,:)));
sixub1 = u(1,:);
sixub2 = u(2,:);
sixrb1 = r(1,:);
sixrb2 = r(2,:);
%%Plot Figure%%
figure %DISPLACEMENT
plot(Time,u(1,:), 'r', 'linewidth',1.5) %Col 1
hold on
plot(Time,u(2,:), 'b', 'linewidth',1.5) %Col 2
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 6 Displacement Response (4B, El Centro)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r', 'linewidth',1.5) %Col 1
hold on
plot(u(2,:),r(2,:), 'b', 'linewidth',1.5) %Col 2
xlabel('Displacment (in)')

```

```
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 6 Restoring Force vs Displacement (4B, El Centro)')
hold off
grid on

%%Section D%% RINALDI RECORD D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-✓
D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-DDDDDD
M = BigM(:, :, 1);
K = BigK(:, :, 1);
C = [0.79281 0.05084; 0.05084 0.55176]; %NOTE DAMPING MATRIX ✓
STAYS THE SAME SINCE K IS THE SAME
ry = Bigry(:, :, 1)';
ddug = SF4*Rinaldi';
dof = DOF;
t = Time2';
dt = dt2;
[r,u, du, ddu] = PART3_NR(M,C,K, ry,t,a,dof,dt,ddug);
umax6d1 = max(abs(u(1,:)));
umax6d2 = max(abs(u(2,:)));
rmax6d1 = max(abs(r(1,:)));
rmax6d2 = max(abs(r(2,:)));
sixud1 = u(1,:);
sixud2 = u(2,:);
sixrd1 = r(1,:);
sixrd2 = r(2,:);
%%Plot Figure%%
figure %DISPLACEMENT
plot(Time2',u(1,:), 'r', 'linewidth', 1.5)
hold on
plot(Time2',u(2,:), 'b', 'linewidth', 1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 6 Displacement Response (4D, Northridge)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r', 'linewidth', 1.5)
hold on
plot(u(2,:),r(2,:), 'b', 'linewidth', 1.5)
```

```
xlabel('Displacment (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 6 Restoring Force vs Displacement (4D, Northridge)')
hold off
grid on
```

```
% Comparison plots for problem 6
```

```
%EL CENTRO
```

```
close all;clc;
subplot(2,1,1)
plot(Time,fourub1,'r','linewidth',1.5)
hold on
plot(Time,sixub1,'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4B', 'Problem 6')
title('Column 1 Response to El Centro')
hold off
grid on
```

```
%DOF 2
```

```
subplot(2,1,2)
plot(Time,fourub2,'r','linewidth',1.5)
hold on
plot(Time,sixub2,'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4B', 'Problem 6')
title('Column 2 Response to El Centro')
hold off
grid on
```

```
%Restoring force
```

```
figure
subplot(2,1,1)
plot(fourub1,fourrb1,'r','linewidth',1.5)
hold on
plot(sixub1,sixrb1,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
```



```
legend('Problem 4B', 'Problem 6')
title('Column 1 Response to El Centro')
hold off
grid on

%DOF 2
subplot(2,1,2)
plot(fourub2,fourrb2,'r','linewidth',1.5)
hold on
plot(sixub2,sixrb2,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4B', 'Problem 6')
title('Column 2 Response to El Centro')
hold off
grid on

%NORTHRIDFEDGE
close all;clc;
subplot(2,1,1)
plot(Time2,fourud1,'r','linewidth',1.5)
hold on
plot(Time2,sixud1,'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4D', 'Problem 6')
title('Column 1 Response to Northridge')
hold off
grid on

%DOF 2
subplot(2,1,2)
plot(Time2,fourud2,'r','linewidth',1.5)
hold on
plot(Time2,sixud2,'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4D', 'Problem 6')
title('Column 2 Response to Northridge')
hold off
grid on
```

```
%Restoring force
```

```
figure
subplot(2,1,1)
plot(fourud1,fourrd1,'r','linewidth',1.5)
hold on
plot(sixud1,sixrd1,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4D', 'Problem 6')
title('Column 1 Response to Northridge')
hold off
grid on
```

```
%DOF 2
```

```
subplot(2,1,2)
plot(fourud2,fourrd2,'r','linewidth',1.5)
hold on
plot(sixud2,sixrd2,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4D', 'Problem 6')
title('Column 2 Response to Northridge')
hold off
grid on
```

```
%%
```

```
%% Part 7
```

```
clc;close all;
```

```
M = wtotal/g;
```

```
K2_part2 = V_eq_part2/(uy*(70/50 + 1));
```

```
K1_part2 = 1.4*K2_part2;
```

```
K2_part2 = K1_part2;
```

```
stiff used
```

```
%Stronger column✓
```

```
ry1_part7 = K1_part2*uy;
```

```
%kip,yield strength✓
```

```
for first column
```

```
ry2_part7 = K2_part2*uy;
```

```
%kip,yield strength✓
```

```
for first column
```

```
wn_part2 = [sqrt(K1_part2/M) sqrt(K2_part2/M)]';
```

```

ry_p2 = [ry1_part7;ry2_part7];

%%PART 1 EL CENTRO %%
M = BigM(:,:,1);
BigK(:,:,1) = [K1_part2      0
               0      K2_part2];
K = BigK(:,:,1);
BigC = zeros(DOF,DOF,1);
%Solve for natural freqencies
[V1,D] = eig(BigK(:,:,1),BigM(:,:,1),"vector");
wn_p1 = sqrt(D);
wn1 = wn_p1(1); wn2 = wn_p1(2);
%SOLVE SYTEM TO GET a0 and a1
wnmat = [1/wn1 wn1;
         1/wn2 wn2];
avals = 2* inv(wnmat)*[xi;xi];
a0 = avals(1);
a1 = avals(2);
BigC(:,:,1) = a0*BigM(:,:,1) + a1*BigK(:,:,1);
C = BigC(:,:,1);
Bigry = zeros(DOF,1,1);
T_fund = wn1 / (2*pi);
Bigry(:,:,1) = [ry1_part7 ry2_part7];
ry = Bigry(:,:,1)';
ddug = SF2*Acc';
dof = DOF;
t = Time;
dt = 0.02; %Given✓

time step in the recording file
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax7b1 = max(abs(u(1,:)));
umax7b2 = max(abs(u(2,:)));
rmax7b1 = max(abs(r(1,:)));
rmax7b2 = max(abs(r(2,:)));

sevenub1 = u(1,:);
sevenub2 = u(2,:);
sevenrb1 = r(1,:);
sevenrb2 = r(2,:);

%%Plot Figure%%

```

```
figure %DISPLACEMENT
plot(Time,u(1,:), 'r','linewidth',1.5) %Col 1
hold on
plot(Time,u(2,:), 'b','linewidth',1.5) %Col 2
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 7 Displacement Response (4B, El Centro)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r','linewidth',1.5) %Col 1
hold on
plot(u(2,:),r(2,:), 'b','linewidth',1.5) %Col 2
xlabel('Displacment (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 7 Restoring Force vs Displacement (4B, El Centro)')
hold off
grid on

%%Section D%% RINALDI RECORD D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-✓
D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-DDDDDD
M = BigM(:, :, 1);
K = BigK(:, :, 1);
C = BigC(:, :, 1);
ry = Bigry(:, :, 1)';
ddug = SF4*Rinaldi';
dof = DOF;
t = Time2';
dt = dt2;
[r,u, du, ddu] = PART3_NR(M,C,K, ry,t,a,dof,dt, ddug);
umax7d1 = max(abs(u(1,:)));
umax7d2 = max(abs(u(2,:)));
rmax7d1 = max(abs(r(1,:)));
rmax7d2 = max(abs(r(2,:)));

sevenud1 = u(1,:);
sevenud2 = u(2,:);
sevenrd1 = r(1,:);
sevenrd2 = r(2,:);
```

```

%%Plot Figure%%
figure %DISPLACEMENT
plot(Time2',u(1,:), 'r', 'linewidth',1.5)
hold on
plot(Time2',u(2,:), 'b', 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 7 Displacement Response (4D, Northridge)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r', 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'b', 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 7 Restoring Force vs Displacement (4D, Northridge)')
hold off
grid on

%COMMENT ON MAX DISP & DUCTILITY FROM 4B and 4D
%%
close all; clc;
%prob 7 plot comparisons
%EL CENTRO
subplot(2,1,1)
plot(Time,fourub1, 'r', 'linewidth',1.5)
hold on
plot(Time,sevenub1, 'b', 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4B', 'Problem 7')
title('Column 1 Response to El Centro')
hold off
grid on

%D0F 2
subplot(2,1,2)
plot(Time,fourub2, 'r', 'linewidth',1.5)

```

```
hold on
plot(Time,sevenub2,'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4B', 'Problem 7')
title('Column 2 Response to El Centro')
hold off
grid on
```

%Restoring force

```
figure
subplot(2,1,1)
plot(fourub1,fourrb1,'r','linewidth',1.5)
hold on
plot(sevenub1,sevenrb1,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4B', 'Problem 6')
title('Column 1 Response to El Centro')
hold off
grid on
```

%DOF 2

```
subplot(2,1,2)
plot(fourub2,fourrb2,'r','linewidth',1.5)
hold on
plot(sevenub2,sevenrb2,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4B', 'Problem 7')
title('Column 2 Response to El Centro')
hold off
grid on
```

%NORTHRIFFDGE

```
close all;clc;
subplot(2,1,1)
plot(Time2,fourud1,'r','linewidth',1.5)
hold on
plot(Time2,sevenud1,'b','linewidth',1.5)
xlabel('Time (sec)')
```

```
ylabel('Displacement (inches)')
legend('Problem 4D', 'Problem 7')
title('Column 1 Response to Northridge')
hold off
grid on
```

```
%DOF 2
```

```
subplot(2,1,2)
plot(Time2,fourud2,'r','linewidth',1.5)
hold on
plot(Time2,sevenud2,'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Problem 4D', 'Problem 7')
title('Column 2 Response to Northridge')
hold off
grid on
```

```
%Restoring force
```

```
figure
subplot(2,1,1)
plot(fourud1,fourrd1,'r','linewidth',1.5)
hold on
plot(sevenud1,sevenrd1,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4D', 'Problem 7')
title('Column 1 Response to Northridge')
hold off
grid on
```

```
%DOF 2
```

```
subplot(2,1,2)
plot(fourud2,fourrd2,'r','linewidth',1.5)
hold on
plot(sevenud2,sevenrd2,'b','linewidth',1.5)
ylabel('F_s (kip)')
xlabel('Displacement (inches)')
legend('Problem 4D', 'Problem 7')
title('Column 2 Response to Northridge')
hold off
```

grid on


```
%% PART 5 (NEW SCRIPT)
```

```
%START WITH PART 1 AND 2 TO GET PARAMETERS
```

```
clear all; close all; clc;
```

```
format shortg
```

```
%%LOAD EL CENTRO DATA%%
```

```
g = 386.09; %Gravity in in/s^2
```

```
Data = load('El_Centro_EQ-Motion.txt');
```

```
Time = Data(:,1);
```

```
Acc = Data(:,2).*g;
```

```
%%LOAD NORTHRIDGE%%
```

```
Data = importdata('Northridge-Rinaldi.txt');
```

```
Rinaldi = Data.data*g;
```

```
Rinaldi = Rinaldi(1:10196);
```

```
dt2 = 0.002; %new dt not same
```

```
Time2max = length(Rinaldi)*dt2;
```

```
Time2 = linspace(0,Time2max,length(Rinaldi));
```

```
%%Given Parameters%%
```

```
DOF =2;
```

```
L1 = 40; L2 = 120; L3 = 20; %ft
```

```
Lgir = 180;
```

```
Ltrib = [Lgir/2 - L3;Lgir/2 - L1];
```

```
rtrib = Ltrib/L2;
```

```
w = 9;
```

```
wtotal = w*Lgir;
```

```
in kips
```

```
M = wtotal/g;
```

```
k*s^2/in
```

```
xi = 0.05;
```

```
and columns)
```

```
%Response spectrum values(Bridge design)
```

```
S_ds = 1.5*g; %units in in/s^2
```

```
S_d1 = g; %units in in/s^2
```

```
T_s = S_d1/S_ds; %Seconds
```

```
T_l = 12; %Seconds
```

```
%Given values for the displacment
```

```
uy    = 3; %yield displacement in inches
umax  = 15; %Max displacement in inches
mu     = umax/uy; %max permitted ductility demand
```

```
%% PART 1 %% REWRITE
```

```
clc;
mu     = umax/uy; %max permitted ductility demand
```

```
%Using Relations for each column
disp('Part 1')
```

```
dt     = 1e-6; %Time increment
Tend   = 14; %end point for period vector
T = [0:dt:Tend]';
n = size(T);
```

```
%Create response spectrum ACC (Figure 2)
```

```
Sa = zeros(n); %initialize ACC
```

```
for i = 1:n
    if T(i) <= T_s
        Sa(i) = S_ds;
    elseif T(i) <= T_l
        Sa(i) = S_d1/T(i);
    else T(i) > T_l;
        Sa(i) = S_d1*T_l/(T(i)^2);
    end
end
```

```
V = wtotal*Sa/(g*mu); %Shear
Kmin = V/uy; %min stiff vector
```

```
%iteration to determine elastic Period
```

```
Trialperiod = zeros(length(T),1);
```

```
dtl = zeros(length(T),1);
```

```
%Elastic period
```

```
for i = 1:length(T)
    Trialperiod(i) = 2*pi/(sqrt(Kmin(i)/(M)));
    dtl(i) = abs(T(i) - Trialperiod(i));
end
```

```
Te_1 = T(find(min(dtl) == dtl,1)); %Location of the elastic✓
```

```
period (index)
fprintf('Telastic = %.3f sec\n',Te_1)

%Stiffness and yield strength via iteration
K_eq = Kmin(find(min(dtt) == dtt,1));
for i = 1:D0F
    ry(i) = rtrib(i) * K_eq * uy;
    K(i) = ry(i)/uy;
end
K_p1 = K;
ry_p1 = ry';
Keq_p1 = K_eq';
fprintf('K1 = %.3f kip/in\n',K(1))
fprintf('K2 = %.3f kip/in\n',K(2))
fprintf('ry1 = %.3f kips\n',ry(1))
fprintf('ry2 = %.3f kips\n',ry(2))
%% PART 2 REWRITE
disp(' ') %Just a divider for the output
disp('PART 2')
D0F = 2;
ti = 0.1;
K_eq1 = ry(1)/umax; % Equivalent Stiffness of one column using✓
relation of ry = K_eq*Umax
EHD1 = (2*ry(1))*2*(umax-uy); %kip-in, area under curve of✓
equivalent linear cycle
xi_eq1 = EHD1/(2*pi*umax^2*K_eq1); %Equivalent damping
K_eq2 = ry(2)/umax; % Equivalent Stiffness of one column using✓
relation of ry = K_eq*Umax
EHD2 = (2*ry(2))*2*(umax-uy); %kip-in, area under curve of✓
equivalent linear cycle
xi_eq2 = EHD2/(2*pi*umax^2*K_eq2); %Equivalent damping
xi_eq = xi_eq2; %Same for both cases
fprintf('xi_eq = %.3f \n',xi_eq)

%Eurocode Amplification
term = sqrt(0.07/(0.02+xi_eq));

%PSEUDO ACC PLOT
figure
plot(T,Sa,'linewidth',1.5)
title('Design Spectrum: ACC')
```

```

xlabel('Period (Sec)')
ylabel('S_A (in/s^2)')
hold on
xline(T_s,'--','T_s');
hold on
xline(T_l,'--','T_l');
grid on
hold off

%Displacmenet Repsonse Spectrum
for i = 1:length(T)
    Sd(i) = Sa(i)*(T(i)/(2*pi))^2; %unit conversion in/s^2 --✓
> in.
    Sd_eq(i) = Sd(i) * term;
end

%Plot DISP Response Spectrum
figure
plot(T,Sd_eq,'r','linewidth',1.75)
hold on
plot(T,Sd,'g','linewidth',1.75)
hold on
yline(umax,'--','U_m_a_x');
title('Design Spectrum: ACC')
xlabel('Period (Sec)')
ylabel('S_d (in)')
legend("\xi_e_q ", "\xi")
xlim([0 5])
grid on

%Determine where the equalivent period is
for i = 1:length(T)
    if round(Sd_eq(i),2) == 15.0
        Tlocation = i;
    end
end

%Determine Teq and Keq and Design Base Shear (Equivalent✓
Values)
T_eq_part2 = T(Tlocation);
K_eq_part2 = (4*pi^2*M/(T_eq_part2^2));

```

```

V_eq_part2 = K_eq_part2*umax;
K_eq_yield= V_eq_part2/uy;
Te_2 = 2*pi/sqrt(K_eq_yield/M);
fprintf('T_eq = %.3f sec \n',T_eq_part2)
fprintf('K_eq = %.3f kip/in \n',K_eq_part2)
fprintf('K_eq_yield = %.3f kip/in \n',K_eq_yield)
fprintf('V_eq = %.3f kip \n',V_eq_part2)
fprintf('T_elastic = %.3f sec \n',Te_2)

```

%Using Relations for each column

```

K2_part2 = V_eq_part2/(uy*(70/50 + 1));
K1_part2 = 1.4*K2_part2;
ry1_part2 = K1_part2*uy; %kip,yield strength✓
for first column
ry2_part2 = K2_part2*uy; %kip,yield strength✓
for first column
wn_part2 = [sqrt(K1_part2/M) sqrt(K2_part2/M)]';
ry_p2 = [ry1_part2;ry2_part2];
fprintf('K1_part2 = %.3f kip/in \n',K1_part2)
fprintf('K2_part2 = %.3f kip/in \n',K2_part2)
fprintf('ry1_part2 = %.3f kip \n',ry1_part2)
fprintf('ry2_part2 = %.3f kip \n',ry2_part2)

```

```
T_fund = (Te_1 + Te_2)/2; %Average of two structures
```

%% PART 5

```

SF1 = 3.6654; %Scale Factor 1 from old script
SF2 = 7.3307; %Scale Factor 2 from old script
SF3 = 0.75379; %Scale Factor 3 from old script
SF4 = 1.5076; %Scale Factor 3 from old script

```

```

clc; close all;
uy_new = 1; %inches
ry1_part2;
ry2_part2;

```

```

K1_part5 = ry1_part2/uy_new;
K2_part5 = ry2_part2/uy_new;

```

%Establish the Rayleigh Damping Matrix

```
BigM = zeros(DOF,DOF,1);
```

```

BigM(:,:,1) = M*[19/36 1/18; 1/18 13/36];

BigK = zeros(DOF,DOF,1);
BigK(:,:,1) = [K1_part5    0
                0          K2_part5];

BigC = zeros(DOF,DOF,1);
%Solve for natural frequencies
[V1,D] = eig(BigK(:,:,1),BigM(:,:,1),"vector");
wn_p1 = sqrt(D);
wn1 = wn_p1(1); wn2 = wn_p1(2);
%SOLVE SYTEM TO GET a0 and a1
wnmat = [1/wn1 wn1;
         1/wn2 wn2];
avals = 2* inv(wnmat)*[xi;xi];
a0 = avals(1);
a1 = avals(2);
BigC(:,:,1) = a0*BigM(:,:,1) + a1*BigK(:,:,1);

Bigry = zeros(DOF,1,1);
Bigry(:,:,1) = ry_p2;
T_fund = wn1 / (2*pi);
fprintf('T_fund = %.3f Sec \n',T_fund)

%Determine the ACC at teh fundamental period
%Determine where the equalivent period is

%%PART 4B %%
clc; close all;
%%Section B%%
a = [1;1];
%%PART 1%%
M = BigM(:,:,1);
K = BigK(:,:,1);
C = BigC(:,:,1);
ry = Bigry(:,:,1)';
ddug = SF2*Acc';
dof = DOF;
t = Time;
dt = 0.02;
time step in the recording file

```

%Given✓

```
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax5b1 = max(abs(u(1,:)));
umax5b2 = max(abs(u(2,:)));
rmax5b1 = max(abs(r(1,:)));
rmax5b2 = max(abs(r(2,:)));
```

```
%%Plot Figure%%
figure %DISPLACEMENT
plot(Time,u(1,:), 'r', 'linewidth',1.5) %Col 1
hold on
plot(Time,u(2,:), 'b', 'linewidth',1.5) %Col 2
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 5 Displacement Response (4B, El Centro)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r', 'linewidth',1.5) %Col 1
hold on
plot(u(2,:),r(2,:), 'b', 'linewidth',1.5) %Col 2
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 5 Restoring Force vs Displacement (4B, El Centro)')
hold off
grid on
```

```
%%Section D%% RINALDI RECORD
```

[illegible]

```
M = BigM(:,: ,1);
K = BigK(:,: ,1);
C = BigC(:,: ,1);
ry = Bigry(:,: ,1)';
ddug = SF4*Rinaldi';
dof = D0F;
t = Time2';
dt = dt2;
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax5d1 = max(abs(u(1,:)));
```

```

umax5d2 = max(abs(u(2,:)));
rmax5d1 = max(abs(r(1,:)));
rmax5d2 = max(abs(r(2,:)));
%%Plot Figure%%
figure %DISPLACEMENT
plot(Time2',u(1,:), 'r', 'linewidth',1.5)
hold on
plot(Time2',u(2,:), 'b', 'linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 5 Displacement Response (4D, Northridge)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r', 'linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'b', 'linewidth',1.5)
xlabel('Displacement (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 5 Restoring Force vs Displacement (4D, Northridge)')
hold off
grid on

%COMMENT ON MAX DISP & DUCTILITY FROM 4B and 4D

%% PART 6
clc;close all;
M = wtotal/g;
%Using Relations for each column
K2_part2 = V_eq_part2/(uy*(70/50 + 1));
K1_part2 = 1.4*K2_part2;
ry1_part2 = K1_part2*uy; %kip,yield strength✓
for first column
ry2_part2 = K2_part2*uy; %kip,yield strength✓
for first column
wn_part2 = [sqrt(K1_part2/M) sqrt(K2_part2/M)]';
ry_p2 = [ry1_part2;ry2_part2];

```


%Reduce ry by 25%

```
ry1_part6 = ry1_part2 * 0.75;
ry2_part6 = ry2_part2 * 0.75;
uy1_p6     = ry1_part6 / K1_part2;
uy2_p6     = ry2_part6 / K2_part2;
```

```
fprintf('uy = %.3f kip/in \n',uy1_p6 )
fprintf('uy = %.3f kip/in \n',uy1_p6 )
fprintf('ry1_part6 = %.3f kip \n',ry1_part6)
fprintf('ry2_part6 = %.3f kip \n',ry2_part6)
```

%%PART 1 EL CENTRO %%

```
M = BigM(:, :, 1);
BigK(:, :, 1) = [K1_part2      0
                 0          K2_part2];
```

```
K = BigK(:, :, 1);
```

```
C = [0.79281 0.05084; 0.05084 0.55176]; %NOTE DAMPING MATRIX✓
```

STAYS THE SAME SINCE K IS THE SAME

```
Bigry(:, :, 1) = [ry1_part6 ry2_part6];
ry = Bigry(:, :, 1)';
ddug = SF2*Acc';
dof = DOF;
t = Time;
dt = 0.02;
```

%Given✓

time step in the recording file

```
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax6b1 = max(abs(u(1,:)));
umax6b2 = max(abs(u(2,:)));
rmax6b1 = max(abs(r(1,:)));
rmax6b2 = max(abs(r(2,:)));
```

%%Plot Figure%%

```
figure %DISPLACEMENT
```

```
plot(Time,u(1,:), 'r', 'linewidth', 1.5) %Col 1
```

```
hold on
```

```
plot(Time,u(2,:), 'b', 'linewidth', 1.5) %Col 2
```

```
xlabel('Time (sec)')
```

```
ylabel('Displacement (inches)')
```

```
legend('Column 1', 'Column 2')
```

```
title('Part 6 Displacement Response (4B, El Centro)')
```

```
hold off
```

```
grid on
```

```
figure %RESTORING FORCE
plot(u(1,:),r(1,:),'r','linewidth',1.5)      %Col 1
hold on
plot(u(2,:),r(2,:),'b','linewidth',1.5)      %Col 2
xlabel('Displacment (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 6 Restoring Force vs Displacement (4B, El Centro)')
hold off
grid on

%%Section D%% RINALDI RECORD D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-DDDDDD
M = BigM(:, :, 1);
K = BigK(:, :, 1);
C = [0.79281 0.05084; 0.05084 0.55176]; %NOTE DAMPING MATRIX✓
STAYS THE SAME SINCE K IS THE SAME
ry = Bigry(:, :, 1)';
ddug = SF4*Rinaldi';
dof = DOF;
t = Time2';
dt = dt2;
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax6d1 = max(abs(u(1,:)));
umax6d2 = max(abs(u(2,:)));
rmax6d1 = max(abs(r(1,:)));
rmax6d2 = max(abs(r(2,:)));

%%Plot Figure%%
figure %DISPLACEMENT
plot(Time2',u(1,:),'r','linewidth',1.5)
hold on
plot(Time2',u(2,:),'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 6 Displacement Response (4D, Northridge)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:),'r','linewidth',1.5)
```

```

hold on
plot(u(2,:),r(2:),'b','linewidth',1.5)
xlabel('Displacment (in)')
ylabel('F_s (Kip)')
legend('Column 1', 'Column 2')
title('Part 6 Restoring Force vs Displacement (4D, Northridge)')
hold off
grid on

```

```
%COMMENT ON MAX DISP & DUCTILITY FROM 4B and 4D
```

```
%% Part 7
```

```
clc;close all;
```

```
M = wtotal/g;
```

```
K2_part2 = V_eq_part2/(uy*(70/50 + 1));
```

```
K1_part2 = 1.4*K2_part2;
```

```
K2_part2 = K1_part2;
```

```
%Stronger column✓
```

```
stiff used
```

```
ry1_part7 = K1_part2*uy;
```

```
%kip,yield strength✓
```

```
for first column
```

```
ry2_part7 = K2_part2*uy;
```

```
%kip,yield strength✓
```

```
for first column
```

```
wn_part2 = [sqrt(K1_part2/M) sqrt(K2_part2/M)]';
```

```
ry_p2 = [ry1_part7;ry2_part7];
```

```
%%PART 1 EL CENTRO %%
```

```
M = BigM(:,:,1);
```

```
BigK(:,:,1) = [K1_part2      0
               0      K2_part2];
```

```
K = BigK(:,:,1);
```

```
BigC = zeros(DOF,DOF,1);
```

```
%Solve for natural freqencies
```

```
[V1,D] = eig(BigK(:,:,1),BigM(:,:,1),"vector");
```

```
wn_p1 = sqrt(D);
```

```
wn1 = wn_p1(1); wn2 = wn_p1(2);
```

```
%SOLVE SYTEM TO GET a0 and a1
```

```
wnmat = [1/wn1 wn1;
```

```
         1/wn2 wn2];
```

```
avals = 2* inv(wnmat)*[xi;xi];
```

```

a0 = avals(1);
a1 = avals(2);
BigC(:, :, 1) = a0*BigM(:, :, 1) + a1*BigK(:, :, 1);
C = BigC(:, :, 1);
Bigry = zeros(DOF, 1, 1);
T_fund = wn1 / (2*pi);
    Bigry(:, :, 1) = [ry1_part7 ry2_part7];
    ry = Bigry(:, :, 1)';
    ddug = SF2*Acc';
    dof = DOF;
    t = Time;
    dt = 0.02; %Given✓
time step in the recording file
[r, u, du, ddu] = PART3_NR(M, C, K, ry, t, a, dof, dt, ddug);
umax7b1 = max(abs(u(1, :)));
umax7b2 = max(abs(u(2, :)));
rmax7b1 = max(abs(r(1, :)));
rmax7b2 = max(abs(r(2, :)));

sevenub1 = u(1, :);
sevenub2 = u(2, :);
sevenrb1 = r(1, :);
sevenrb2 = r(2, :);

%%Plot Figure%%
figure %DISPLACEMENT
plot(Time, u(1, :), 'r', 'linewidth', 1.5) %Col 1
hold on
plot(Time, u(2, :), 'b', 'linewidth', 1.5) %Col 2
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 7 Displacement Response (4B, El Centro)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1, :), r(1, :), 'r', 'linewidth', 1.5) %Col 1
hold on
plot(u(2, :), r(2, :), 'b', 'linewidth', 1.5) %Col 2
xlabel('Displacement (in)')
ylabel('F_s (Kip)')

```

```
legend('Column 1', 'Column 2')
title('Part 7 Restoring Force vs Displacement (4B, El Centro)')
hold off
grid on

%%Section D%% RINALDI RECORD D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-✓
D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-DDDDDD
M = BigM(:,:,1);
K = BigK(:,:,1);
C = BigC(:,:,1);
ry = Bigry(:,:,1)';
ddug = SF4*Rinaldi';
dof = DOF;
t = Time2';
dt = dt2;
[r,u, du, ddu] = PART3_NR(M,C,K,ry,t,a,dof,dt,ddug);
umax7d1 = max(abs(u(1,:)));
umax7d2 = max(abs(u(2,:)));
rmax7d1 = max(abs(r(1,:)));
rmax7d2 = max(abs(r(2,:)));

sevenud1 = u(1,:);
sevenud2 = u(2,:);
sevenrd1 = r(1,:);
sevenrd2 = r(2,:);
%%Plot Figure%%
figure %DISPLACEMENT
plot(Time2',u(1,:), 'r','linewidth',1.5)
hold on
plot(Time2',u(2,:), 'b','linewidth',1.5)
xlabel('Time (sec)')
ylabel('Displacement (inches)')
legend('Column 1', 'Column 2')
title('Part 7 Displacement Response (4D, Northridge)')
hold off
grid on
figure %RESTORING FORCE
plot(u(1,:),r(1,:), 'r','linewidth',1.5)
hold on
plot(u(2,:),r(2,:), 'b','linewidth',1.5)
xlabel('Displacment (in)')
```

```
ylabel('F_s (Kip)')  
legend('Column 1', 'Column 2')  
title('Part 7 Restoring Force vs Displacement (4D, Northridge)')  
hold off  
grid on
```

```
%COMMENT ON MAX DISP & DUCTILITY FROM 4B and 4D
```